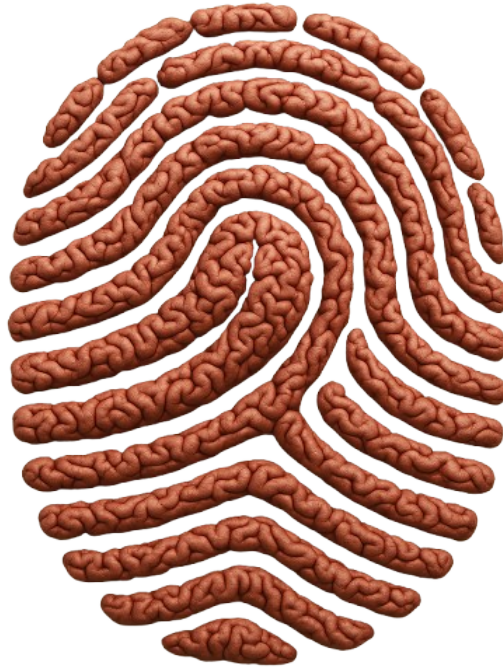




# On the optimization of diffusion-enhanced MRF sequences through flip angle and phase train design with comparative phase modulation strategies

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# 1 Abstract

Magnetic Resonance Fingerprinting (MRF) enables simultaneous estimation of multiple tissue properties within a single scan. Recent advances have extended this framework to incorporate diffusion sensitivity, allowing joint mapping of relaxation and microstructural features. This work presents a principled optimization strategy for designing diffusion-enhanced MRF sequences that robustly encode  $T_1$ ,  $T_2$ , proton density ( $M_0$ ), mean diffusivity (MD), and fractional anisotropy (FA). The optimization is grounded in the Cramér–Rao Lower Bound (CRLB), enabling direct minimization of parameter estimation uncertainty. Flip angle and phase modulation trains are co-optimized via smooth B-spline parametrizations across ensembles of physiologically plausible tissue models, ensuring robustness to tissue variability. Several distinct phase modulation strategies are systematically evaluated. Additionally, we investigate the inclusion of an inversion pulse module, comprising an inversion pulse and its own optimized flip angle and phase modulation trains, as a replacement for the first two diffusion modules in the baseline sequence. This modification aims to increase contrast and enhance encoding performance early in the sequence. Simulations show that the optimized sequences substantially reduce normalized CRLBs relative to baseline methods. Among the tested phase strategies, second-order modulation yields the best trade-off between parameter sensitivity and robustness across tissue types. It was found that the inversion preparation module degrades encoding efficiency, underscoring the importance of data-driven over heuristic design. These findings establish that joint flip angle and phase modulation optimization offers a powerful framework for robust, diffusion-sensitive MRF protocol development. By unifying relaxation and diffusion mapping in a single, time-efficient acquisition, the proposed approach advances the potential for clinically viable quantitative MRI with enhanced diagnostic value.

## 2 Introduction

Magnetic Resonance Imaging (**MRI**) forms one of the most versatile imaging technique in clinical settings. However, MRI is predominantly qualitative, which complicates the extraction of quantitative information from the images. Consequentially, medical doctors and clinicians rely foremost on visual assessment for the detection of pathologies. Though the contrast in these images is usually governed by  $T_1$  and  $T_2$  [1] properties of the imaged tissues, these parameters are usually not directly measured in most of the standardized protocols. The measured quantity is the magnitude of MR signal at a particular time point, after the evolution of the imaged system under a specific set of parameters of the sequence.

A significant portion of recent advancements in the field of MRI has been directed towards the development of image acquisition techniques to obtain quantitative maps. This quest has led to the development of techniques such as (but not limited to) quantitative susceptibility mapping (QSM) [2], diffusion tensor imaging (DTI) [3] and magnetic resonance elastography (MRE) [4]. These provide quantitative imaging information regarding parameters such as magnetic susceptibility, diffusion properties and tissue stiffness. A notable recent advancement introduced by Ma et al [5] in the field of quantitative MRI has been the development of the **Magnetic Resonance Fingerprinting** (MRF) [6] framework. In this approach, dynamic variation of sequence parameters results in signals for each voxel that act as a fingerprint characterizing underlying tissue properties. Each such signal can be matched to entries in a pre-generated dictionary based on Bloch equations. This approach enables the simultaneous acquisition of multiple quantitative maps in one scan, featuring parameters such as  $T_1$ ,  $T_2$  and  $M_0$ . More recently, significant effort has been put into the integration of **Diffusion-Weighted Imaging** (DWI) into the MRF framework. These works respectively comprised rapid high-resolution whole-brain imaging featuring Apparent Diffusion Coefficient ( $ADC$ ) [7] & **Fractional Anisotropy** ( $FA$ ) [8] mapping.

Building upon these developments, the aim of this project was to contribute to the ongoing refinement of the MRF framework through the optimization of an acquisition scheme. This is achieved through the introduction of an optimization strategy aiming to improve precision in parameter estimation. More precisely, this is achieved by leveraging the Cramér-Rao Lower Bounds [9] (CRLB). The latter is the theoretical limit on the variance of any unbiased estimator of a certain parameter. This provides a method to evaluate the information content of a certain acquisition sequence with regards to a certain parameter (examples:  $T_1$ ,  $T_2$ ,  $M_0$ ). The incorporation of the CRLBs into the optimization objective is motivated by the project seeking to design an MRF sequence with a strong emphasis on its sensitivity to all the studied underlying tissue parameters.

To realize this, a flexible sequence parameterization was adopted using B-spline [10] representations of the flip angle train, allowing the optimization algorithm to explore a rich and smooth space of candidate sequences as initially proposed by Crafts et al [11]. This approach was particularly suited to encode complex temporal modulation patterns while maintaining a manageable number of degrees of freedom. The optimization was carried out using the JAX framework, which enabled fast automatic differentiation and CPU-accelerated computation of gradients through the Bloch simulation and CRLB evaluation pipeline. The cost function was constructed by computing Fisher information matrices across a representative ensemble of tissue parameter sets drawn from a physiological range. The trace of the inverse Fisher information matrix, weighted according to a predefined distribution of expected tissue types, was used as the scalar objective to minimize. This ensured that the optimized sequence performed robustly across a broader spectrum of tissue characteristics (here: white matter, gray matter and cerebro-spinal fluid, according to the values as reported by Helenius et Al [12]) rather than being tuned to a single tissue type.

In addition to flip angle modulation, this work implemented time-varying phase modulation as an additional degree of freedom in the sequence optimization. Though applied in some recent applications [13] [14] but traditionally underutilized in MRF optimization workflows, phase modulation optimization has the potential to diversify signal evolutions and improve parameter encoding, particularly when aiming to disentangle correlated signal behaviors, which constitute a problem for accurate dictionary matching. The inclusion of this component requires the extension of the Bloch simulation model to handle arbitrary phase cycling, and carefully accounting for its impact on the complex signal trajectory. Initial experiments revealed that the combined use of flip angle

and phase modulation resulted in more diversified and information-rich signal palette, consequentially improving the conditioning of the dictionary-matching step and the related inverse problem.

This study also investigated the inclusion of an inversion pulse module, with its own flip angle and phase train, as a replacement for the first two diffusion modules of the baseline sequence. This aimed to boost early contrast and encoding performance, but results showed that it reduced the overall efficiency.

Furthermore, recognizing the growing interest in diffusion-MRF and the potential of its robust integration in the framework, the project lays groundwork for integrating diffusion sensitivity into the optimization pipeline. While previous similar studies [15] [16] primarily focused on relaxation parameter encoding ( $T_1$ ,  $T_2$ ), in this study, simulations were conducted incorporating diffusion effects, parameterized by **Mean Diffusivity** [17] ( $MD$ ) and Fractional Anisotropy ( $FA$ ). The long-term goal is to develop a unified sequence capable of jointly estimating relaxation and diffusion parameters in a time-efficient manner. Such a sequence would be of considerable clinical relevance, offering a consolidated acquisition protocol that preserves spatial resolution while reducing scan time thereby minimizing patient discomfort.

In summary, this project contributes a novel optimization framework for MRF sequence design rooted in information-theoretic principles, augmented by B-spline parametrization and phase modulation. The results demonstrate that optimization of both amplitude and phase characteristics of the sequence yields improvements in parameter estimation precision. These findings pave the way for more advanced MRF protocols that integrate diffusion encoding and other biophysical contrasts, ultimately moving MRI towards more comprehensive and reliable quantitative imaging.

## 3 Theoretical background

### 3.1 MRI

Magnetic resonance imaging relies on an intrinsic property of hydrogen atoms, called the **spin** angular momentum [18] [19]. The possession of an odd mass number or odd atomic number by a nucleus results in the latter being characterized by a non-zero nuclear spin  $I$  [20] [21]. The presence of the non-zero value results in a magnetic moment. The linear relation between nuclear spin  $I$  and magnetic moment  $\mu$  is governed by the gyromagnetic ratio  $\gamma$ , which is nucleus specific.

Most applications of MRI employ the precession of hydrogen atoms to generate images. This preference is justified by several factors, one of which is hydrogen's high gyromagnetic ratio ( $\gamma = 2.675 \cdot 10^9 \frac{\text{rad}}{\text{T}\cdot\text{s}}$ ) compared to other biologically relevant nuclei such as  $^{23}\text{Na}$ ,  $^{35}\text{Cl}$ , or  $^{39}\text{K}$  [22].

In a magnetically unperturbed system, i.e. characterized by the absence of added or induced magnetic fields, magnetic moments of hydrogen atoms are randomly oriented. This results in there being no net magnetization. The addition of an external magnetic field  $B_0$  results in spins aligning either parallel or antiparallel to the magnetic field, with a slight excess of spins ending up in the antiparallel (i.e. lower energy) state [23] [24]. The unbalance in the distribution of the alignments results in a non-zero net magnetization  $M_0$  in the direction of  $B_0$  [25]. The magnitude of  $M_0$  is prescribed by Curie's law [26] [27], increasing in a linear fashion with  $B_0$ .

The applied magnetic field is not constant in time.

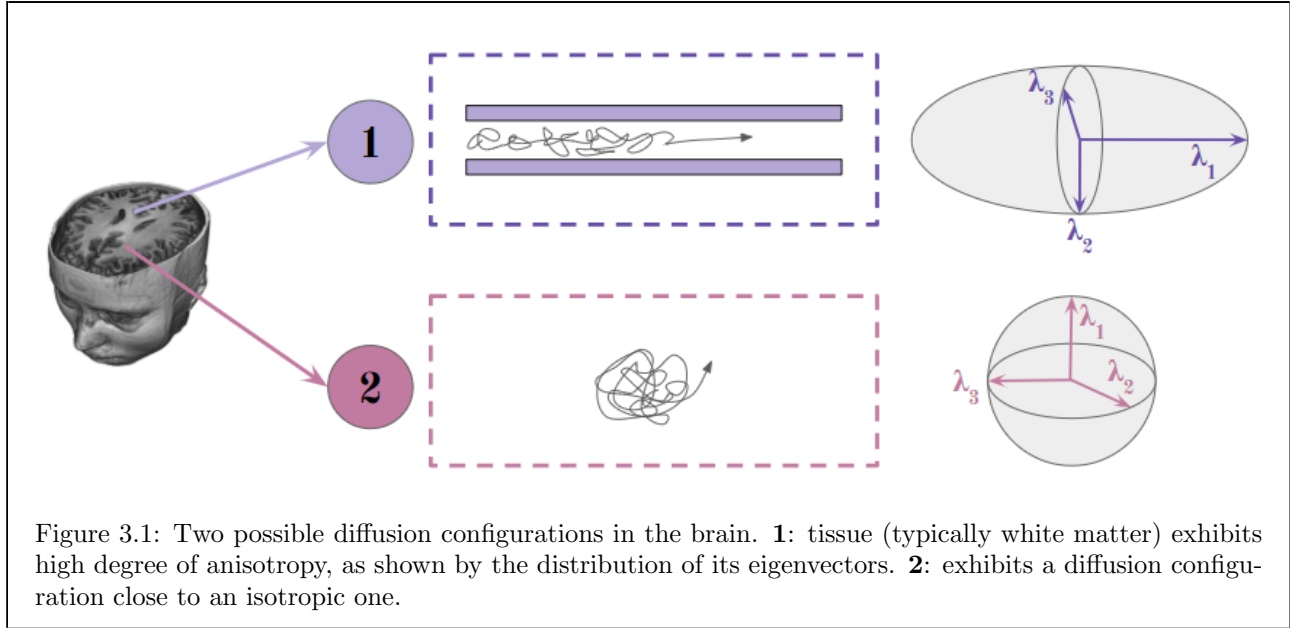
A typical MRI system features a superconducting magnet resulting in a constant magnetic field  $B_0$  in a direction referred to as the **longitudinal plane**. For the purpose of changing the direction of the magnetization, a secondary magnetic field can be introduced. It is often given the notation  $B_1$ . This addition to the initial magnetic field  $B_0$  results in the rotation of the magnetization vector towards the **transverse plane** from the longitudinal plane. The angle of rotation of the magnetization ( $\alpha$ , **flip angle**) is a function of the time the  $B_1$  RF field has been applied and its strength according to Equation 3.1 [28].

$$\alpha = \gamma \int_0^t B_1(t) dt \quad (3.1)$$

After its flip away from its initial point of equilibrium, the magnetization starts to precess around the main magnetic field  $B_0$ . This precession in the transversal direction can be measured from a nearby receiver coil thanks to its induction of current as described by Faraday's law of induction [29]. This enables the reconstruction of images.

The latter is achieved through the application of spatially varying magnetic fields [30] (i.e. gradients) along different axes. This results the hydrogen spins to have spatially-varying the resonance frequencies, which enables the decoding of spatial information in the frequency and the phase of the received signal [31].

The contrast in these simple images will in most cases be dictated by 2 relaxation mechanisms. After the excitation of the spins by an RF pulse, the longitudinal magnetization relaxes back to its rest situation from the excited state at a rate represented by the  $T_1$  relaxation time [32]. This  $T_1$  relaxation time varies per tissue (for example, the  $T_1$  times for fat are shorter than those for water) and is the basis of the contrast in the  $T_1$ -weighted images. These basic MR images depict the magnitude of the magnetization that has recovered in the longitudinal direction at a chosen time after the excitation pulse [33]. The transverse relaxation, represented by the  $T_2$  time, measures the rate at which the transverse magnetization of the protons decays after the application of the RF pulse [34].  $T_2$  relaxation is a direct consequence of interactions of the protons with the surrounding environment which results in the dephasing of the magnetic moments and consequentially a reduction of the net magnetization in the transverse plane. In reality however, the transverse magnetization undergoes a decay at a higher speed than what would be predicted through solely  $T_2$ . This faster rate is modeled by  $T_2^*$ , and includes the effects of inhomogeneities in the magnetic field [35].



**3.1.1 Diffusion tensor imaging** Diffusion is defined as the movement of something, usually driven by a concentration gradient [36]. Particles moving freely in all directions are described by isotropic diffusion (in the brain, typically, water in gray matter) while a more restrained, unidirectional propagation pattern reflects anisotropic diffusion (in the brain, typically, water in white matter). This is further illustrated in Figure 3.1. Diffusion tensor imaging (DTI) is a method proposed by Basser et al. [37] [38] for describing this anisotropic diffusion behavior in fibrous tissues through a multivariate normal distribution [39]. In the latter, the diffusion tensor  $\mathbf{D}$  represents the covariance of the possible diffusion displacements, normalized by the diffusion time. This is defined as the interval during which water molecules can diffuse within the tissue while being imaged. The simplest implementation of a so-called diffusion preparation scheme is depicted in Figure 3.2. It makes use of 2 gradient pulses on each side of the  $180^\circ$  refocusing pulse. The first of these pulses serves as a way to dephase the magnetization while the other one serves as a rephasing tool. Such a configuration will result in the enabling of the measurement of the diffusion, as for molecular motion in the direction of the applied gradient, the magnetization is sensitized to motion [40], and consequentially the diffusion weighted signal is affected. For immobile molecules, the phase changes introduced by both gradients will cancel out, resulting in null attenuation of the signal as a result of diffusion.

**3.1.2 Magnetic resonance fingerprinting** Standard quantitative MRI sequences generally only allow the extraction of a single parameter (such as but not limited to:  $T_1$ ,  $T_2$ , proton density, ADC) at the time, through the repetition of the sequence with a single setting changed. Instead of sequentially imaging the different parameters, magnetic resonance fingerprinting provides an efficient method to address this shortcoming, and measure multiple parameters within a single sequence. This is done through varying certain sequence parameters (namely flip angle, FA, and repetition time, TR) throughout the sequence, which results in unique signals for many possible combination of the underlying parameters [41], i.e. "fingerprints". In order to determine the underlying tissue parameters, a so-called dictionary of theoretical signal evolutions is generated using Bloch equations [42], for every expected combination of the tissue properties. The measured fingerprint (obtained through the acquisition of many images) of each voxel can then be associated to certain tissue properties by matching the measured signal with the dictionary entry that reassembles it the most. The latter process is robust to undersampling artifacts, which enables faster acquisition. This strengthens the efficiency of the MRF framework [43] [44]. A schematic representation of the image is shown in Figure 3.3.

**3.1.3 Cramér-Rao lower bounds** For the purpose of optimizing the MRF sequences in the presence of noise, Cramer-Rao lower bounds [45] constitute a very robust metric. The latter represents the lowest possible variance of a parameter estimate assuming an unbiased maximum likelihood estimator [46]. In effect, this can be used as the optimization objective in a variety of different MRI sequences [47] [48]. This method can be employed provided there is no bias in the estimators [49]. Analytically, the Cramer-Rao Lower bound of the

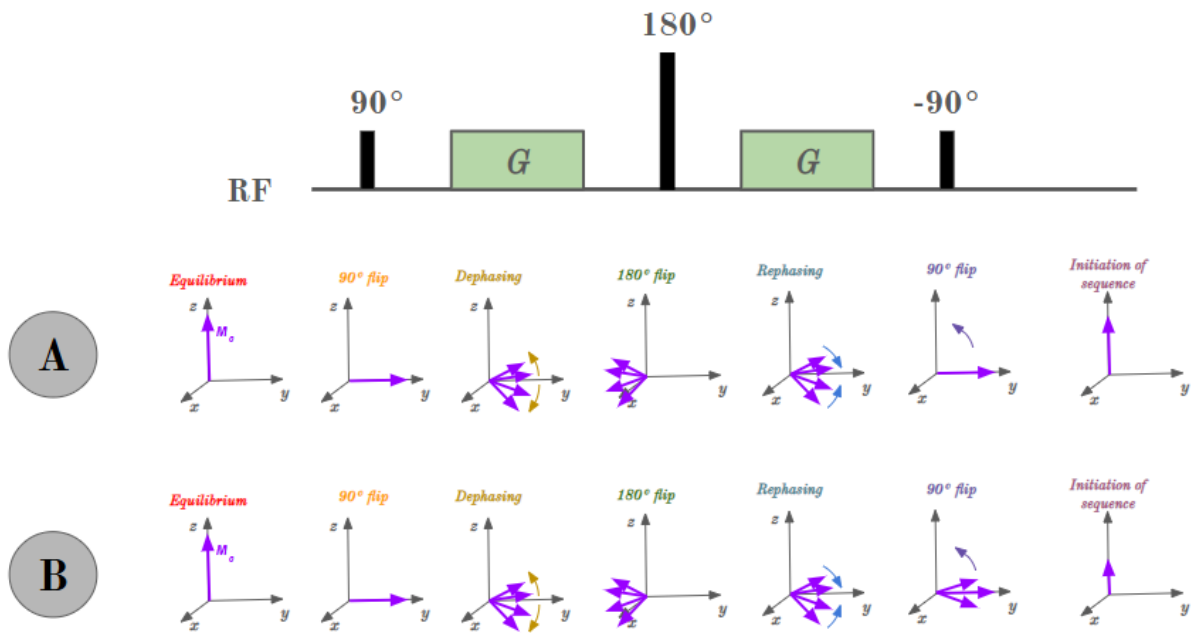


Figure 3.2: Schematic representation of a diffusion module in two different scenarios. **A:** There is no diffusion and therefore no net movement. The spins rephase perfectly thanks to the second gradient and therefore most of the initial magnetization is restored ( $T_1$  and  $T_2$  effects still act on the magnetization as per usual). **B:** As a consequence of diffusion, spins undergo random motion. This results in their accumulation of different phase shifts while in the presence of the gradients. Consequentially, they do not rephase completely after the second gradient, which leads to a loss of signal. In addition, relaxation effects also act on the magnetization as usual.

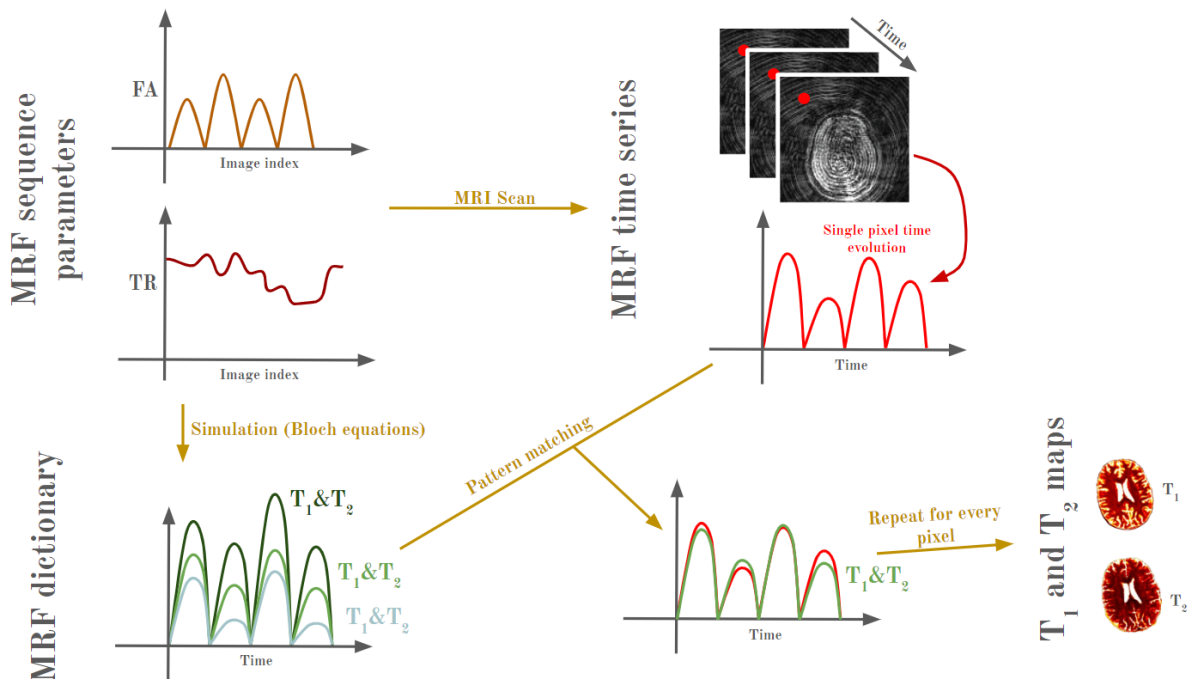


Figure 3.3: Schematic representation of the MRF workflow.

$i$ th parameter estimate  $\hat{\theta}_i$  in an array of unknown parameters can be presented as follows:

$$Var(\hat{\theta}_i) \geq |V(\theta)|_{i,i} \quad (3.2)$$

$$V(\theta) = I^{-1}(\theta) \quad (3.3)$$

Where  $I$  represents the Fisher information matrix [50]. The latter quantifies the information content of measurements.

**3.1.4 Extended phase graph formalism** For a large part of MRI sequences, the analytical expression for the steady state and the magnetization is quite simple to derive, and the optimization process may thus be implemented in a relatively straightforward way. On the other hand, sequences that do not offer this simple analytical expression, namely MRF sequences, require different approaches. Previously, Zhao et al [16] modelled the signal at every echo as the succession of affine transformations [51], the latter governed by Bloch equations [42]. The Cramer-Rao lower bounds were computed by dividing the individual voxels into separate isochromats (i.e. small groups of spins which share the same frequency of resonance [52]). Generally, for MRF sequences, these are simulated using the Extended Phase Graph (EPG) formalism. This method consists of the decomposition of the magnetization in a certain voxel into a basis of both transverse and longitudinal states to model this voxel [53]. The states in these bases can be represented as helicoids reflecting the orientation of isochromats within this particular voxel. During a certain TR, the precession resulting from gradients and off-resonances will result in the magnetization in the transverse plane to evolve to a different state. By making use of simple building blocks, the signal coming from complex sequences can be simulated [54]. The concepts of EPG formalism, along with Bloch equations and Cramer-Rao bound optimization were employed by Lee et al [46]. Their study focused on the discussion of the advantages of the implementation of automatic differentiation within the framework of sequence optimization based on CRLB. Automatic differentiation is not synonymous with numerical differentiation which estimates derivatives based on values of a function at different points. Automatic differentiation employs the principle that every function, no matter how complicated, may be broken down into a sequence represented by elementary operations (addition, multiplication, etc) and elementary functions ( $e^x$ ,  $\sin$ ,  $\cos$ , etc). Consequentially, by applying the chain rule, partial derivatives of any order can easily be computed [55].

## 4 Methods

### 4.1 Pulse sequence description

**4.1.1 Base Sequence** A custom Magnetic Resonance Fingerprinting (MRF) sequence simulation was used, designed to model both relaxation and diffusion effects via an extended phase graph (EPG) formalism. The signal evolution is governed by user-defined flip angle and phase modulation trains, while incorporating anisotropic diffusion using a generalized diffusion tensor  $\mathbf{D}$ . The sequence structure is detailed below.

#### Sequence structure.

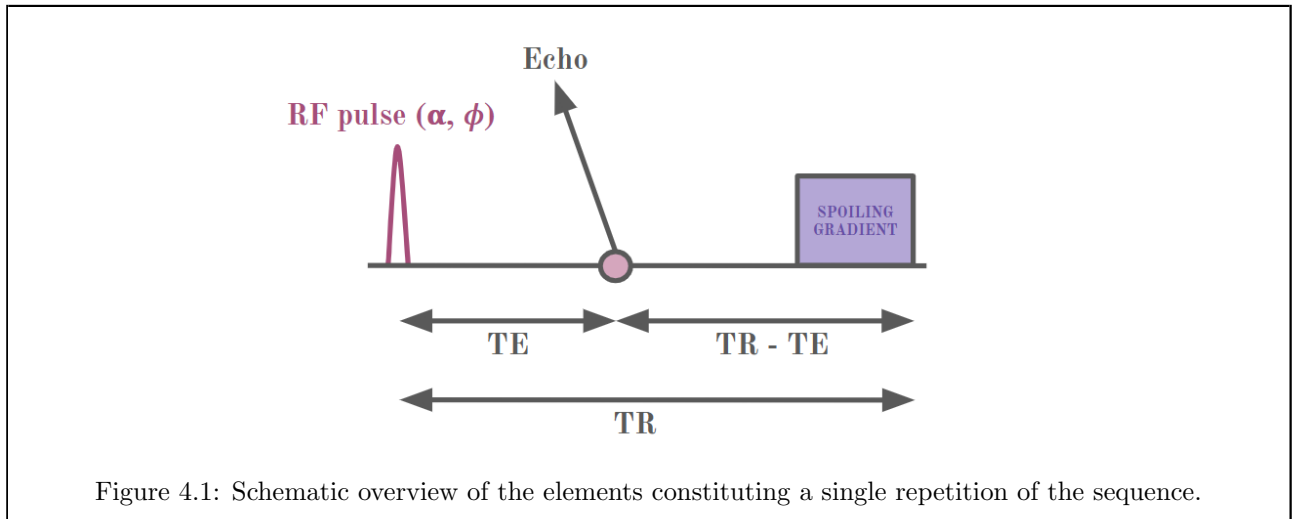
The core of the sequence consists of  $N$  excitations, each with a specified flip angle  $\alpha_i$  and RF phase  $\phi_i$ , followed by relaxation and diffusion evolution steps. Each repetition includes:

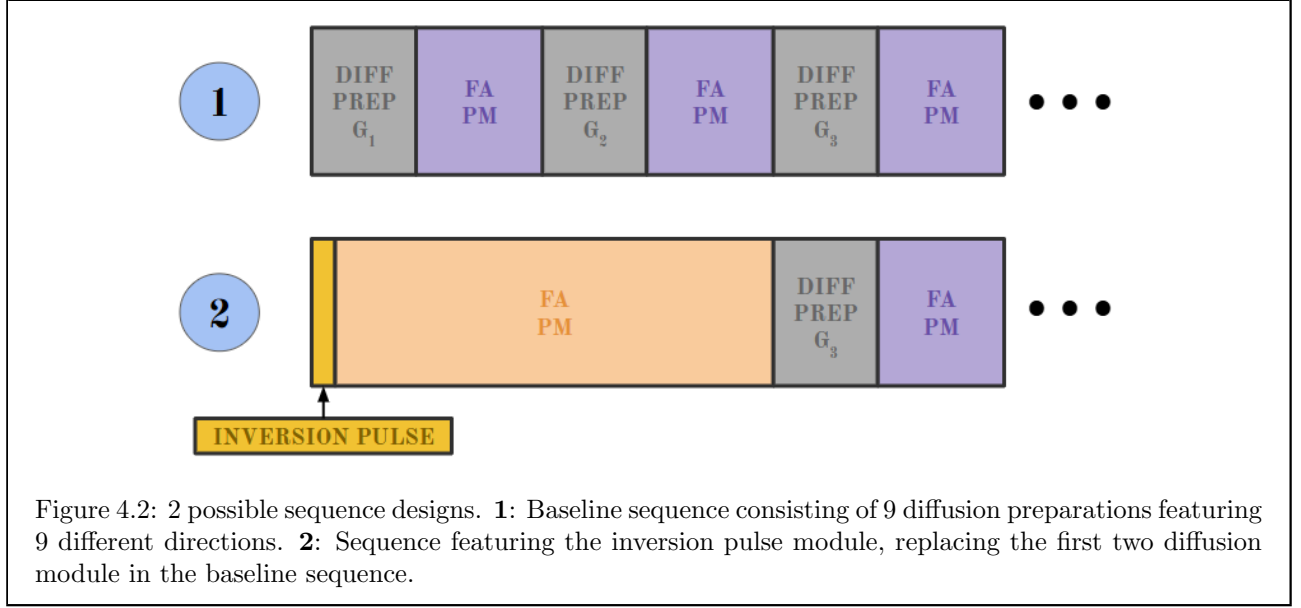
1. An RF pulse with flip angle  $\alpha_i$  and phase  $\phi_i$ ,
2. Evolution during an echo time  $TE = 4$  ms with gradients off,
3. Relaxation and diffusion evolution during  $TR - TE = 11$  ms, combined with the application of a spoiling gradient.

As such, the total repetition time is fixed at  $TR = 15$  ms. The RF pulse train  $\{\alpha_i\}_{i=1}^N$  and  $\{\phi_i\}_{i=1}^N$  is user-specified and optimized for parameter encoding. The evolution of the magnetization vector is modeled using EPG formalism extended to include diffusion using EPG matrix operators. A schematic representation of one such repetition is displayed in Figure 4.1.

#### Diffusion encoding.

To capture anisotropic diffusion, the sequence cycles through  $n_g$  diffusion preparations, described in subsection 3.1.1, featuring different diffusion encoding directions. The baseline sequence is defined as one having  $n_g = 9$ . The flip angle and phase trains subsequent to each of the diffusion preparations in the baseline sequence do not vary between each diffusion encoding direction. A sequence with a prepended inversion pulse





module is also considered. The inversion pulse module features its own flip angle and phase trains, replacing two diffusion modules from the baseline sequence, resulting in  $n_g = 7$ . The performance of the sequence incorporating the inversion preparation module is evaluated in comparison to the baseline sequence composed of diffusion preparation modules only. A schematic comparison of the two sequences is shown in Figure 4.2. The diffusion evolution is governed by a user-defined diffusion tensor  $\mathbf{D} \in \mathbb{R}^{3 \times 3}$ . Maximum gradient strength in one of the 3 orthogonal directions ( $x, y, z$ ) is fixated at a constant effective  $b$ -value of  $1000 \frac{s}{mm^2}$  and is not varied during the sequence.

### Parameter encoding.

This sequence is designed to simultaneously encode:

1.  $T_1$  and  $T_2$  relaxation via relaxation-induced state transitions,
2. Mean diffusivity (MD) and fractional anisotropy (FA) via the gradient-dependent diffusion terms in the EPG model,
3. Equilibrium magnetization  $M_0$ .

### Fixed vs. variable elements.

1. **Fixed:** TE = 4 ms, TR = 15 ms, number of EPG states = 5, gradient strength ( $b = 1000 \frac{s}{mm^2}$ ).
2. **User-defined (variable):**
  - (a) Flip angle train  $\{\alpha_i\}$ ,
  - (b) RF phase train  $\{\phi_i\}$ ,
  - (c) Number of gradient directions  $n_g$  (default: 9),
  - (d) Diffusion tensor  $\mathbf{D}$ ,
  - (e) Whether to prepend the inversion.

### Comparison to conventional MRF.

Compared to standard MRF implementations like FISP-MRF [56] and bSSFP-MRF [57], this method introduces several important distinctions:

1. **Incorporation of anisotropic diffusion** sensitivity due to bipolar gradient inclusion. The effect of the diffusion is modeled via tensor-based EPG evolution, enabling simultaneous mapping of MD, FA,  $T_1$ , and  $T_2$ ,

2. **Gradient cycling** to probe microstructural anisotropy in several orientations,
3. **Optional preparatory inversion block** for improved tissue contrast, especially for the relaxation parameters, with flip angle and phase trains optimized separately from the flip angle and phase trains subsequent to diffusion preparations.

#### Output.

The simulated output is a complex-valued signal time series (real and imaginary components), optionally undersampled in time (not applicable in the case of this study) to reflect practical acquisition constraints. This output forms the basis for dictionary matching or parameter estimation algorithms.

## 4.2 B-Spline Parameterization

**4.2.1 Rationale** The parametrization of the flip angle and phase trains using a B-spline model and optimizing its coefficients instead of optimizing an unprocessed signal offers many advantages:

1. **Reduced dimensionality of the optimization problem:** Approximating the flip angle and phase trains using a spline representation reduces the dimensionality of the problem, replacing the explicit specification of one flip angle and phase value per time point with a smaller set of spline coefficients. This yields a more compact parameterization while preserving signal fidelity within a controlled approximation error.
2. **Smoothness control:** A B-spline is, by definition, constructed from piecewise polynomial functions that are smoothly joined at specified locations known as knots [58]. An  $n$ th-degree B-spline yields a function that is  $C^{n-1}$  continuous, provided the knots are simple (i.e., have multiplicity one) [59]. This intrinsic continuity makes B-splines particularly well-suited for enforcing smoothness in curve representations. In optimization problems, such as those involving flip angle trains, this smoothness is advantageous, as derivative constraints can be naturally and efficiently imposed. By contrast, enforcing such constraints directly on each parameter (e.g., individual flip angles) would be significantly more complex.

#### 4.2.2 Implementation Details

1. **Spline order:** fixed at 3. This ensures  $C^2$  continuity which augments the accuracy of the B-spline approximation in some of the phase optimization method.
2. **Knot placement strategy:** the knots are spread uniformly in the time domain. This is to ensure even coverage of the signal, which drastically simplifies the implementation and also avoids bias.
3. **Boundary conditions:** no boundary conditions were imposed on the B-spline approximation of the signal.

## 4.3 Optimization Framework

**4.3.1 Objective Function** To optimize the design of a Magnetic Resonance Fingerprinting (MRF) sequence for robust quantitative imaging, a combined objective function is defined that simultaneously accounts for the encoding efficiency of multiple tissue and diffusion parameters. The function evaluates the performance of a candidate flip angle (FA) and phase train by quantifying the precision of parameter estimation across a physiologically relevant range of tissue properties.

#### Objective formulation.

The objective function is constructed around the Cramér-Rao Lower Bound (CRLB). For a given tissue parameter (i.e. particular  $T_1$  and  $T_2$  values and diffusion tensor) indexed by  $i$ , the Fisher Information Matrix (FIM),  $\mathbf{FIM}_i$ , is computed from the simulated MRF signal response to applied MRF scheme. The signal simulation is performed using an extended phase graph (EPG) formalism that incorporates relaxation and diffusion effects,

including direction-dependent diffusion tensors. The scalar loss associated with configuration  $i$  is defined as the weighted sum accross all parameters of normalized Cramér-Rao lower bounds, as shown in Equation 4.1

$$\text{nCRLB}_i = \left( \sum_{j=1}^P w_j \cdot \left( \frac{[\mathbf{FIM}_i^{-1}]_{jj}}{p_{ij}^2} \right)^{\frac{\gamma}{2}} \right)^{\frac{1}{\gamma}} \quad \text{with} \quad \mathbf{FIM}_i = \frac{1}{\sigma^2} \sum_{n=1}^N \mathbf{J}_{n,i}^T(\mathbf{j}) \mathbf{J}_{n,i}(\mathbf{j}), \quad (4.1)$$

Where:

1.  $\mathbf{J}_n(\mathbf{j})$  the Jacobian computed with regards to parameter with index  $j$ ,
2.  $\sigma^2$  the specified noise level fixated at  $\sigma^2 = 33$  dB in this study,
3.  $\mathbf{FIM}_i$  the Fisher Information Matrix for a certain tissue configuration  $i$ ,
4.  $[\mathbf{FIM}_i^{-1}]_{jj}$  is the variance bound for parameter with index  $j$ ,
5.  $p_{ij}$  is the true value of parameter with index  $j$  in configuration  $i$ ,
6.  $w_j$  is a weight that prioritizes accuracy of specific parameters (e.g.,  $T_1$ ,  $T_2$ ,  $M_0$ , mean diffusivity (MD), or fractional anisotropy (FA)),
7.  $\gamma \in \{1, 2\}$  selects between an L1- and L2-type aggregation across parameters.

#### Aggregate loss across parameter configurations.

To enforce robustness across a diverse set of tissues, the total loss function aggregates the CRLB contributions over all simulated parameter configurations:

$$\mathcal{L}_{\text{total}} = \sum_{i=1}^N w_i^{(2)} \cdot \text{nCRLB}_i, \quad (4.2)$$

Where  $w_i^{(2)}$  is an additional scalar weight applied to each configuration, allowing emphasis on clinically relevant tissues or specific anatomical regions. It is defined such as  $\sum_{i=1}^N w_i^{(2)} = 1$ . In the case of this study  $w_i^{(2)} = \frac{1}{3}$  for all  $i \in \{1, 2, 3\}$ , as we intend to model three tissues: grey matter, white matter and CSF.

#### Parameter domains and sensitivity targets.

The parameter space includes scalar quantities ( $T_1, T_2, M_0$ ) and second-order diffusion tensors  $\mathbf{D}_i$ . From each diffusion tensor, scalar diffusion metrics, namely mean diffusivity (MD) and fractional anisotropy (FA), are extracted and included in the set of parameters whose encoding sensitivity is targeted. The objective thus explicitly accounts for both relaxation and microstructural features.

#### Optimization rationale.

Minimizing  $\mathcal{L}_{\text{total}}$  leads to flip angle and phase modulation trains that enhance encoding efficiency across a wide range of tissue types. Unlike conventional MRF sequence design approaches that rely on intuition or heuristic metrics, this formulation is grounded in estimation theory and explicitly minimizes uncertainty bounds on the recovered parameters. As such, it provides a principled and quantitative framework for sequence optimization under realistic noise and tissue variability conditions.

### 4.3.2 Constraints

#### Flip angle constraints

The optimization process enforces several constraints to ensure physically realistic and clinically viable flip angle trains.

1. **Hard bounds on flip angles:** The optimized flip angle trains are constrained to remain within a lower bound ( $\alpha_n \geq \frac{\pi}{18}$ ) and an upper bound ( $\alpha_n \leq \frac{\pi}{3}$ ). Additionally, slow variation of the flip angle is encouraged via constraints on the finite differences of its derivatives:

$$|\alpha_{n+1} - \alpha_n| \leq \frac{\pi}{180}.$$

2. **Bound constraints on optimization variables:** All B-spline coefficients are bounded within a symmetric interval (e.g.,  $[-100, 100]$ ) to prevent numerical instability or unphysical solutions.

### Phase modulation constraints

No explicit constraints are imposed on the phase modulation train. This allows the optimizer full flexibility to exploit arbitrary phase patterns, which may be beneficial for encoding efficiency or artifact suppression.

### 4.3.3 Phase Modulation Optimization

#### Selection of the initial phase

Several candidate phase modulation functions were evaluated to minimize the initial total Cramér-Rao Lower Bounds (CRLB) for the parameter estimation problem. The considered phase modulation schemes include, each parameterized by a parameter  $j \in [0, 1]$  (which modulates a certain characteristic of the chosen functions):

1. **Quadratic Phase Modulation:**

$$\phi_n = \begin{cases} \phi_{n-1} + 2n, & 0 < n < Nj \\ \phi_{n-1} - 2n, & Nj + 1 < n < N \end{cases} \quad \text{with } \phi_0 = 0,$$

where  $n = 1, 2, \dots, N - 1$  is the time index.

2. **Linear Phase Modulation:**

$$\phi_n = \begin{cases} \phi_{n-1} + 10, & n < Nj \\ \phi_{n-1} - 10, & n > Nj + 1 \\ \phi_{n-1}, & \text{otherwise} \end{cases} \quad \text{with } \phi_0 = 0.$$

3. **Sinusoidal Phase Modulation:**

$$\phi_n = A \sin \left( 2\pi \left( \frac{n}{N} + j \right) \right), \quad n = 0, 1, \dots, N - 1.$$

4. **Alternating Phase Modulation:**

$$\phi_n = \phi_{n-1} + (-1)^n \cdot A \cdot j, \quad \phi_0 = 0, \quad n = 1, \dots, N - 1.$$

Here,  $N$  is the total number of points (length of the flip angle array), and the phase  $\phi_n$  is expressed in degrees and converted to radians for computations.

For each method and fraction  $j$ , the total Cramér-Rao Lower Bound (CRLB) sum

$$S_j = \sum_p \text{CRLB}_p(\phi_j)$$

was computed over all parameters  $p$  of interest.

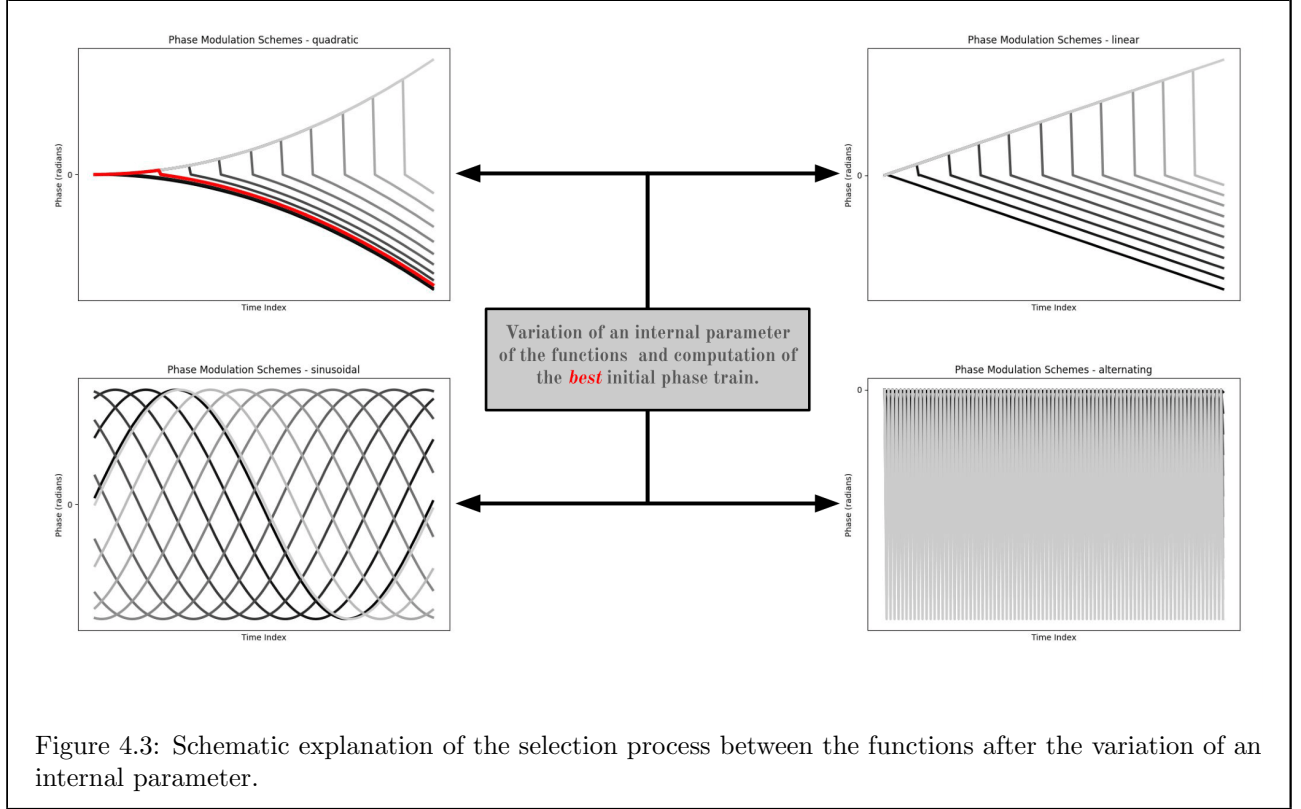
The best fraction  $j^*$  for each method minimizes the total CRLB:

$$j^* = \arg \min_j S_j.$$

Finally, the initial phase modulation method  $\phi^*(t)$  was chosen as the one with the minimal total CRLB across all methods and fractions:

$$m^*, j^* = \arg \min_{m, j} S_j^{(m)}.$$

This selection procedure ensures the best initial phase modulation guess for subsequent optimization and is furthermore illustrated in Figure 4.3



### Phase Modulation Optimization Methods

Let  $t$  denote the evaluation points and  $\phi(t)$  the phase modulation function. The following phase optimization methods are considered:

1. **"No Phase Modulation" Method** (in results figures: *"No phase modulation"*)

Phase modulation is fixed to zero for all  $t$ :

$$\phi(t) = 0.$$

This corresponds to optimizing only the flip angle train without any phase modulation.

2. **"Free Form" Method** (in results figures: *"Free form"*)

The phase modulation  $\phi(t)$  is represented directly as a B-spline expansion without integration:

$$\phi(t) = \sum_j c_{\phi,j} B_j(t),$$

where  $c_{\phi,j}$  are the coefficients and  $B_j(t)$  the B-spline basis functions. This provides full flexibility to  $\phi(t)$  within the smoothness constraints of the B-spline basis.

3. **"Second order modulation" Method** (in results figures: *"Quadratic"*)

The phase modulation  $\phi(t)$  is constructed by integrating a B-spline representation of its second derivative, and adding an affine (constant + linear) component:

$$\phi(t) = \int_0^t \int_0^\tau \text{BSpline}_{c_\phi}(s) ds^2 + C_1 t + C_0,$$

Where  $\text{BSpline}_{c_\phi}(t)$  denotes a B-spline basis expansion with coefficients  $c_\phi$ , which control the second derivative of  $\phi(t)$ . The constants  $C_0$  and  $C_1$  represent the initial offset and linear slope of the phase train. They are computed from the initial phase modulation pattern and remain fixed throughout the optimization. This formulation ensures that  $\phi(t)$  has a smooth second derivative governed by the B-spline, similar to the approach of Fuderer et al. [13].

#### 4. "Internal Parameter Modulation" Method (in results figures: "Quadratic malleable")

Instead of constructing  $\phi(t)$  directly in time, this method optimizes the internal parameters of the phase train generation function itself. If  $\phi(t)$  is defined via a parameterized transformation  $\phi(t; \theta)$ , where  $\theta$  represents internal control variables (e.g., frequency, amplitude, curvature), then optimization is performed directly over  $\theta$ :

$$\phi(t) = \mathcal{F}(t; \theta),$$

where  $\mathcal{F}$  is a fixed functional form whose behavior is steered by the parameter vector  $\theta$ . This approach decouples optimization from spline basis representation.

A summary of these methods is shown in Figure 4.4.

**4.3.4 Numerical optimization details** Methods were implemented in Python. The optimization procedure is based on the `scipy.optimize.minimize` routine using the Sequential Least Squares Quadratic Programming (SLSQP) backend. This method was selected due to its capability to handle both equality and inequality constraints, which are essential to enforce physically and technically meaningful limits on the flip angle and phase modulation train. The full implementation of the proposed optimization schemes and the used simulation tools can be found at [https://github.com/ErikVi/MRF\\_diffusion\\_optimization](https://github.com/ErikVi/MRF_diffusion_optimization) [60]. Results were visualized with the Matplotlib Python library [61].

## 4.4 Optimized sequence validation

To evaluate the performance of the optimized sequence under realistic conditions, synthetic phantoms with varying relaxation and diffusion parameters can be employed. Due to time constraints, satisfactory reconstructed maps could not be obtained. Consequently, validation results are not presented or discussed in the results section. Detailed information regarding the validation procedure is provided in the section A.1.

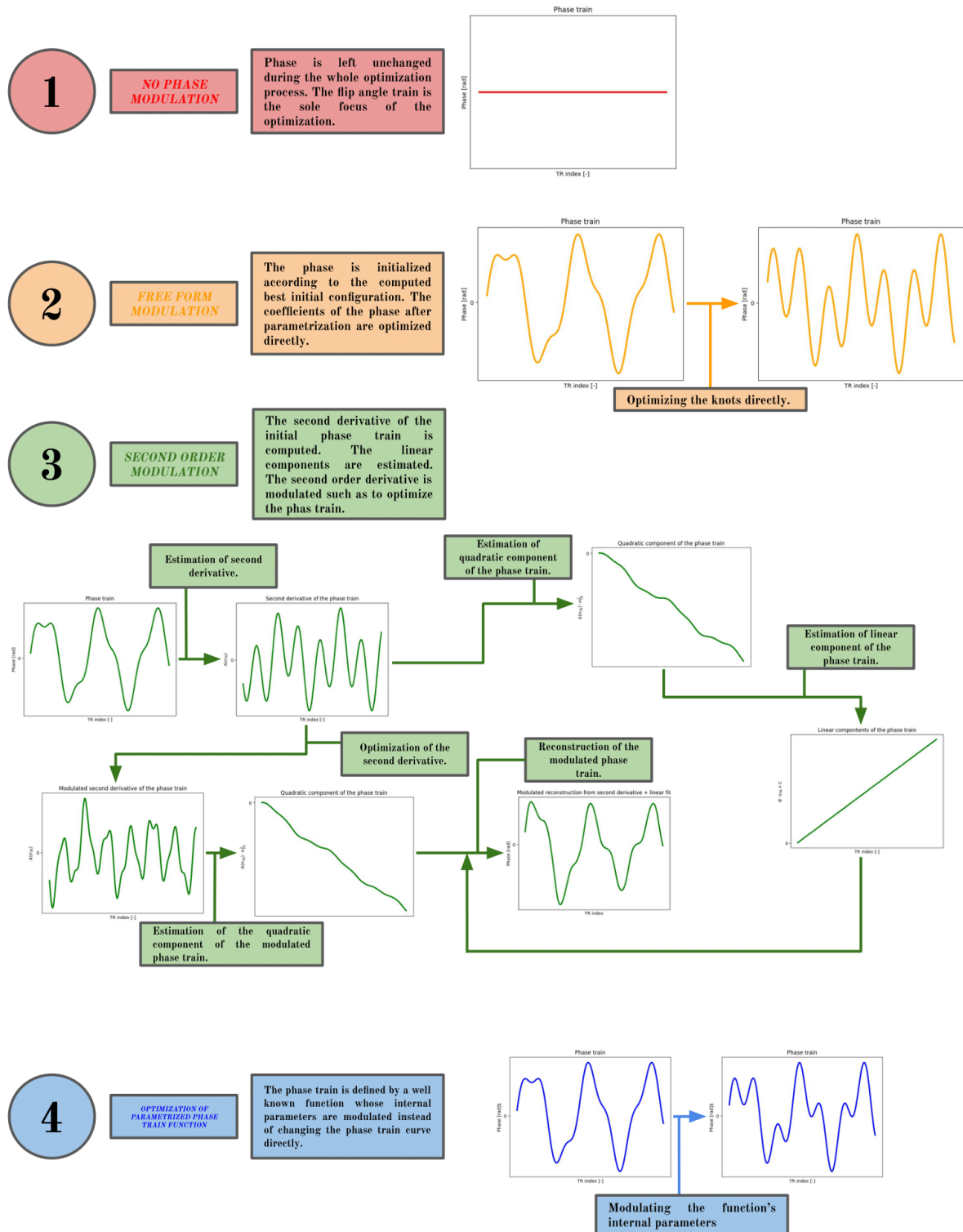


Figure 4.4: Schematic comparison of the 4 considered phase optimization methods.

## 4.5 Methods summary

| Component                            | Details                                                                                            | Options / Notes                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>1. Parameterization</b>           |                                                                                                    |                                                                                                |
| Flip Angle Train<br>Phase Modulation | B-spline (order 3), uniform knots<br>B-spline (free form or 2nd order integrated)                  | No boundary conditions, $C^2$ smooth<br>Also tested: constant, sinusoidal, alternating, linear |
| <b>2. Constraints</b>                |                                                                                                    |                                                                                                |
| FA bounds                            | $\alpha \in [\pi/18, \pi/3]$                                                                       | Inequality constraint via spline interpolation                                                 |
| FA smoothness                        | $ \alpha_{n+1} - \alpha_n  \leq \pi/180$                                                           | Enforced on FA differences                                                                     |
| Phase bounds                         | No constraints                                                                                     | Fully free for optimizer                                                                       |
| Spline coeff. bounds                 | $[-100, 100]$                                                                                      | Prevents instabilities                                                                         |
| <b>3. Objective Function</b>         |                                                                                                    |                                                                                                |
| Per tissue config.                   | Normalized<br>$\left( \sum_j w_j \cdot \sqrt{[FIM^{-1}]_{jj}/p_{ij}^2} \right)^\gamma \text{CRLB}$ | $\gamma = 1$ (L1) or 2 (L2)                                                                    |
| Total loss                           | $L = \sum_i w_i^{(2)} \cdot \text{nCRLB}_i$                                                        | Weighted over all tissue configs                                                               |
| <b>4. Optimization Settings</b>      |                                                                                                    |                                                                                                |
| Method                               | SLSQP via<br><code>scipy.optimize.minimize</code>                                                  | Handles inequality constraints                                                                 |
| Gradient                             | JAX autodiff (forward mode)                                                                        | JIT-compiled, vectorized                                                                       |
| Initialization                       | Predefined FA and optimal phase guess                                                              | Based on CRLB minimization                                                                     |
| Convergence                          | Tolerance $\leq 10^{-2}$ or max iterations                                                         | Manual <code>StopIteration</code> supported                                                    |
| <b>5. Numerical Details</b>          |                                                                                                    |                                                                                                |
| Signal Simulation                    | EPG formalism (5 states) + diffusion                                                               | Includes anisotropic diffusion tensors                                                         |
| FA Train Lengths                     | 170 Flip angles per module                                                                         | Swept in outer loop                                                                            |
| Phase Methods                        | No phase, Free-form, 2nd-order, or direct modulation of internal parameters                        | Compared in ablation                                                                           |

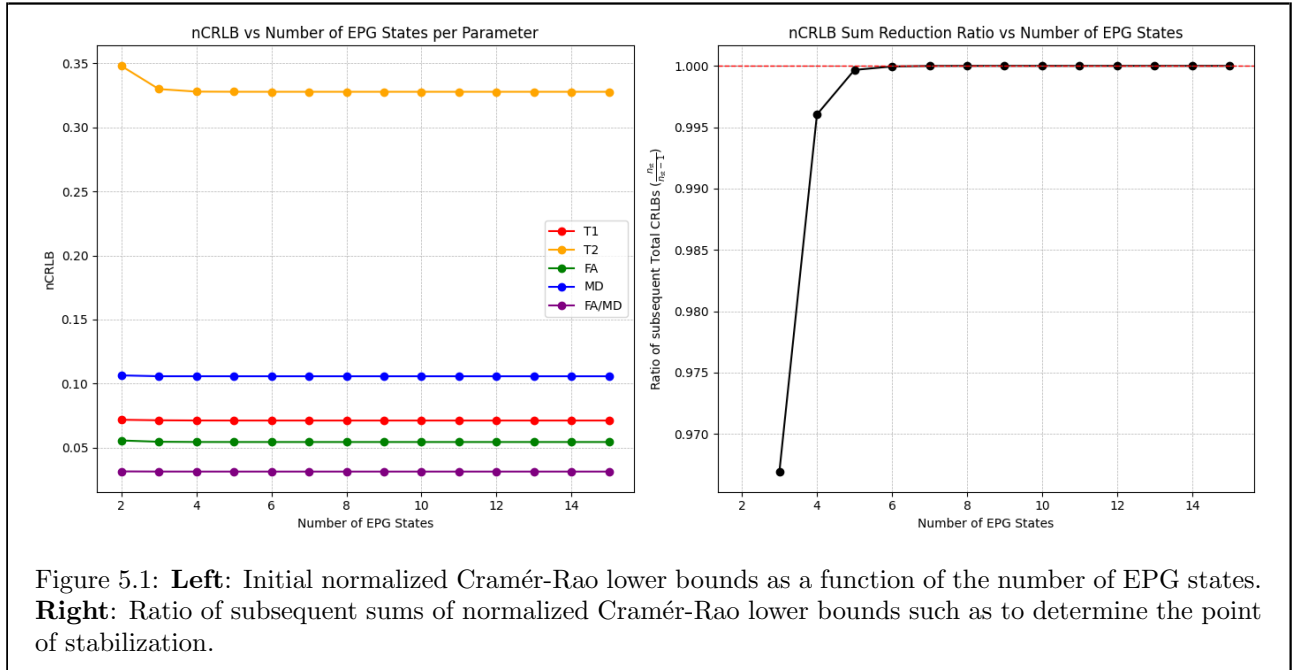
## 5 Results

This chapter presents the results of the optimization and simulation framework developed in this work. First, the choice of knots and the number of EPG states is justified. After that, the effect of different phase modulation strategies on the normalized Cramér-Rao lower bounds (nCRLBs) is analyzed for all estimated parameters. Subsequently, we evaluate different combinations of flip angle trains, phase modulation strategies, and inversion preparation schemes. Finally, we present the signal evolutions obtained from the best-performing protocol and assess their information content qualitatively. The summary of this section is provided in section 5.6.

### 5.1 Effect of the number of EPG States

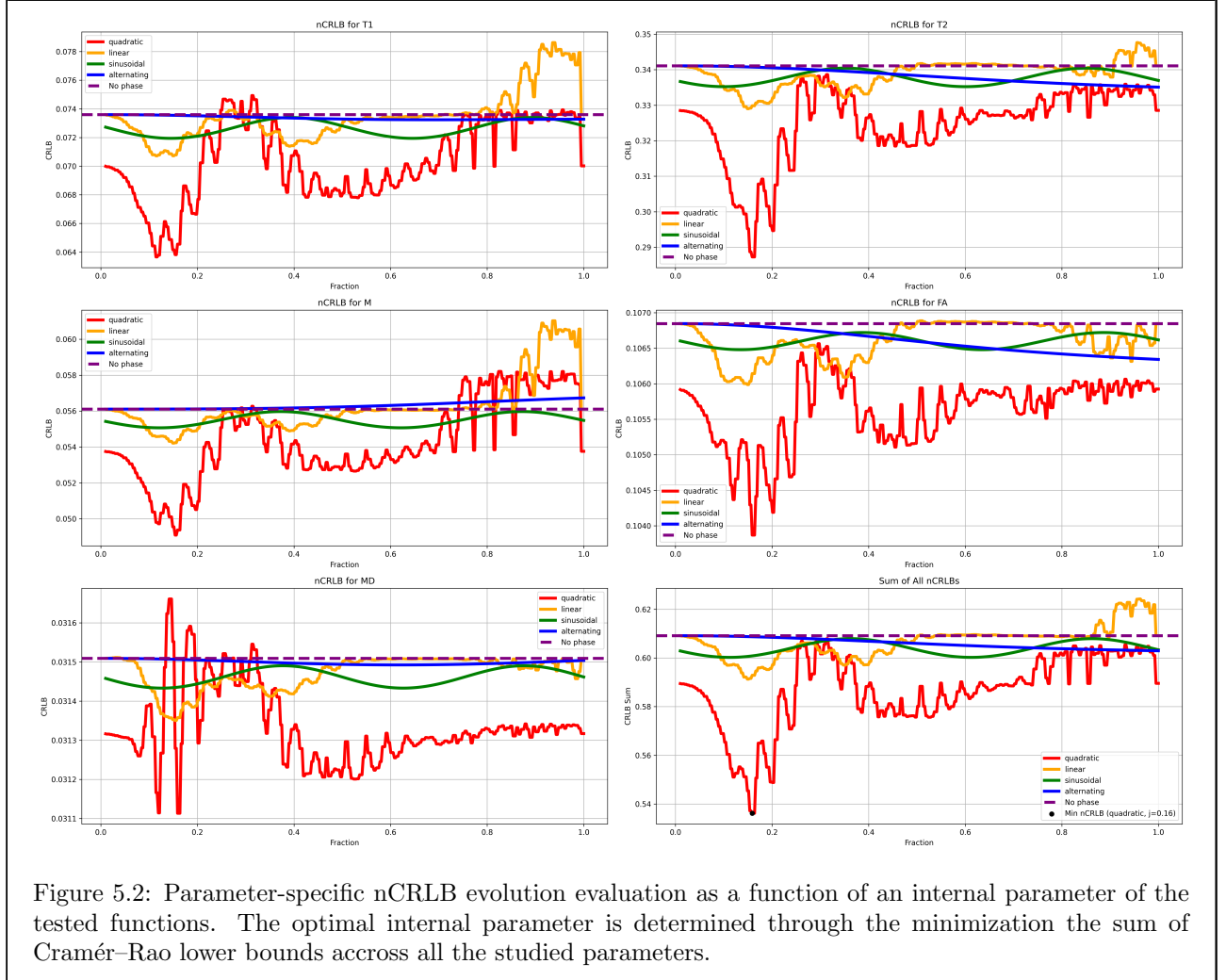
Accurate modeling of the magnetization dynamics under repeated RF pulses requires a sufficient number of EPG states. However, computational burden scales rapidly with the number of tracked states. The trade-off between accuracy and computational cost is evaluated by plotting the total nCRLB (summed across parameters and dictionary entries) as a function of the number of EPG states. The results shown in Figure 5.1 demonstrate that the nCRLB values for all parameters stabilize rapidly as the number of EPG states increases. Beyond approximately **5 EPG states**, the computed Normalized Cramér-Rao lower bounds for the initial configurations are governed by close to no changes at all. In particular, the nCRLBs for  $T_1$ ,  $T_2$ , and FA show only marginal improvements beyond this point, and those for PD and MD remain virtually unchanged.

This convergence behavior suggests that increasing the number of EPG states beyond five does not yield any significant benefit in estimation accuracy. Consequently, for the purposes of this study, using five EPG states represents an effective trade-off between computational efficiency and model fidelity, as it ensures stable and reliable parameter estimation while minimizing simulation time and memory usage.



## 5.2 Initial trains

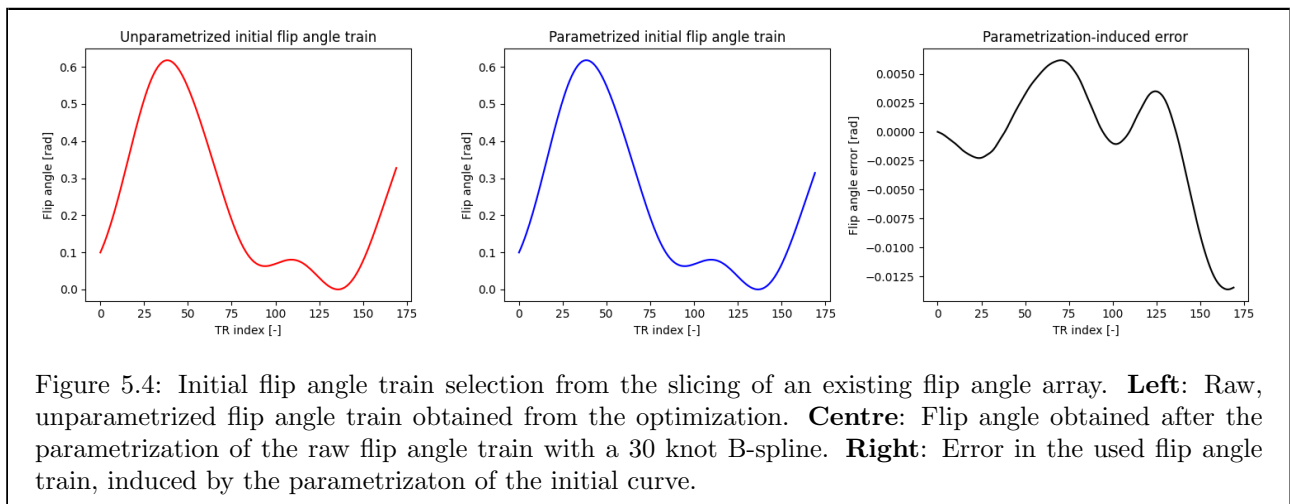
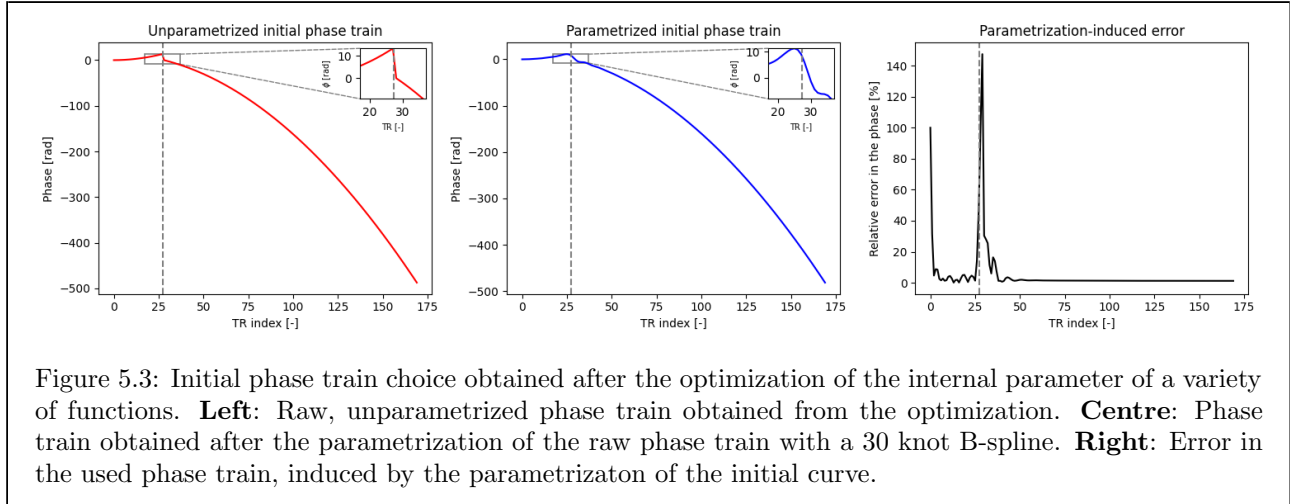
**5.2.1 Initial phase modulation train** As described in subsubsection 4.3.3, a number of different phase trains are tested for a given range of internal parameters. The method and internal parameter combination, resulting in a certain phase train, are then selected and used as an initialization for the phase optimization process. The results of the optimization are displayed in Figure 5.2. The resulting best initial phase train,



its parametrization and the error in the parametrization are displayed in Figure 5.3. The discontinuity at TR index 27 (in Figure 5.3, marked with a vertical gray striped line) is the largest source of error between the raw and the parametrized phase train, as the relative uncertainty reaches a climax in that point. This stems from the parametrization of a discontinuous function with a B-Spline (which is by definition continuous), and errors of this magnitude were expected for the particular segment.

**5.2.2 Initial flip angle train** The initial flip angle train used in this work is a truncated version of the optimized MRF flip angle train proposed by Sommer et al. [62] in their study on the optimal pattern length for MRF sequences. This sequence was selected as a high-performing baseline and served as the initialization for the subsequent optimization procedure. The employed initialization is shown in Figure 5.4.

Contrary to the phase train, the B-spline captures the curve of the raw flip angle train with ease, resulting in very small errors in the angles. In Figure 5.4, only relative error is shown in the rightmost figure, as the small values of the angles around TR index 135 result in a numerical instability leading to large relative errors. This larger error in reality does not reflect the quality of the approximation.



### 5.3 Effect of the number of B-spline knots

The parametrization of the to-be-optimized curves using B-spline is justified by the reduction of the dimensionality of the optimization problem. Consequentially, it is of great interest to minimize the number of knots defining the flip angle and phase trains. As shown in Figure 5.5, strong variations in parameter-specific nCRLBs halt at about 20 knots.

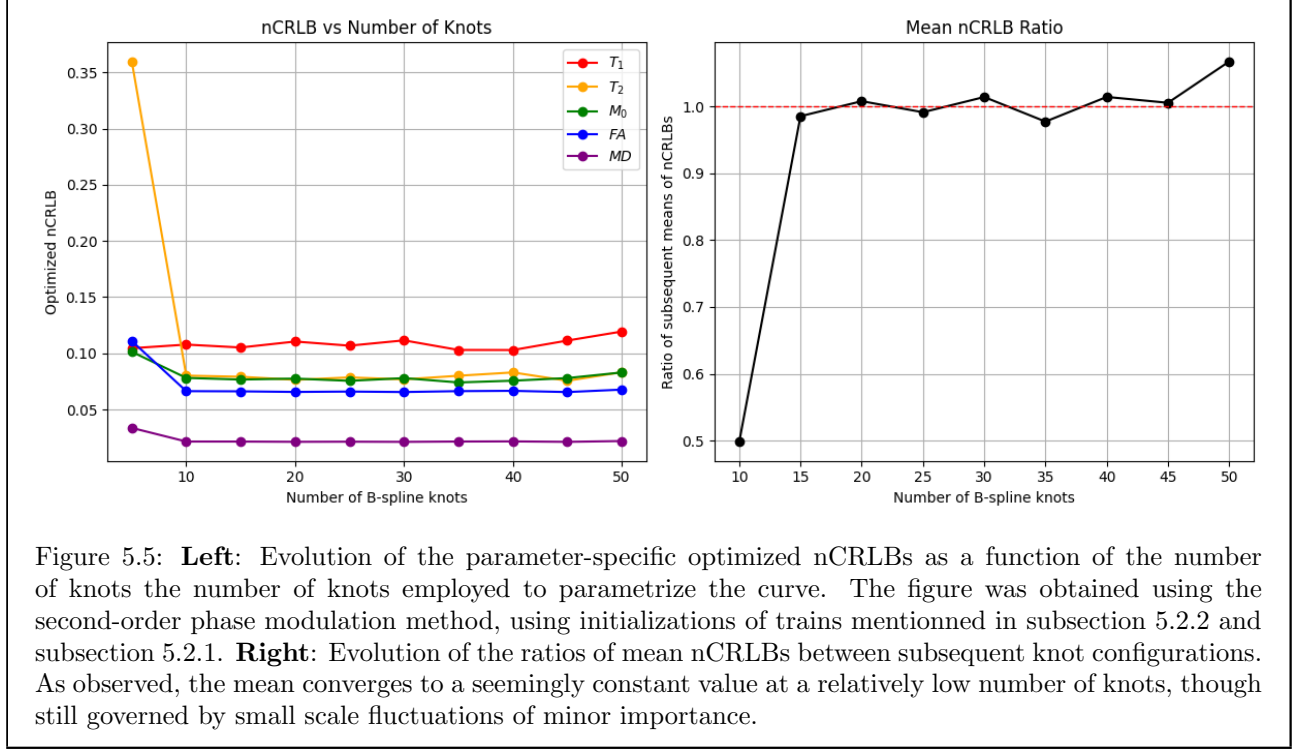


Figure 5.5: **Left:** Evolution of the parameter-specific optimized nCRLBs as a function of the number of knots the number of knots employed to parametrize the curve. The figure was obtained using the second-order phase modulation method, using initializations of trains mentioned in subsection 5.2.2 and subsection 5.2.1. **Right:** Evolution of the ratios of mean nCRLBs between subsequent knot configurations. As observed, the mean converges to a seemingly constant value at a relatively low number of knots, though still governed by small scale fluctuations of minor importance.

## 5.4 Optimization results

**5.4.1 Optimization of the optional inversion pulse preparation flip angle and phase trains** The optional inversion pulse preparation module (light orange block in Figure 4.2) flip angle and phase trains are optimized using the same methods and optimization parameters as the flip and phase trains following diffusion preparations. Comparing the mean normalized Cramér-Rao lower bounds across all methods, as shown in Figure 5.6, shows the (relatively small) superiority of the second-order phase modulation method (*Quadratic* in Figure 5.6).

The resulting flip angle and phase trains are displayed in Figure 5.7. The optimized flip angle train can be described by a localized transient increase centered at about TR index 30, followed by a section where it is close to baseline (equal to  $\frac{\pi}{18}$  rad). Subsequently, a sharp rise at the maximum possible speed occurs ( $\frac{\pi}{180}$  rad per TR), and the angles then seem to plateau at the maximum possible flip angle value (equal to  $\frac{\pi}{3}$  rad). This result may be interpreted as the optimization yielding a two-stage sequence, with the first (low flip angles) one being governed by low-perturbation dynamics, while the second one (high flip angles) clearly reflects stronger excitation conditions.

**5.4.2 Cramér-Rao lower bounds for every optimization method/sequence type** Such as to evaluate the performance of all possible phase optimization methods, combined with the optional inclusion of the inversion pulse preparation module and the possibility to switch to a L2 type cost function, all the parameter-specific Cramér-Rao lower bounds are compared. The following sections detail this comparison for a total of  $4 \cdot 2 \cdot 2 = 16$  optimizations (4 phase optimization methods, 2 possible sequence designs, and 2 cost function orders). Though the per-parameter normalized Cramér-Rao lower bounds optimization results carry much information about the internal workings of the optimization process, the main metric used to evaluate the quality of a specific combination of optimization parameters is the mean nCRLBs of across all parameters, and

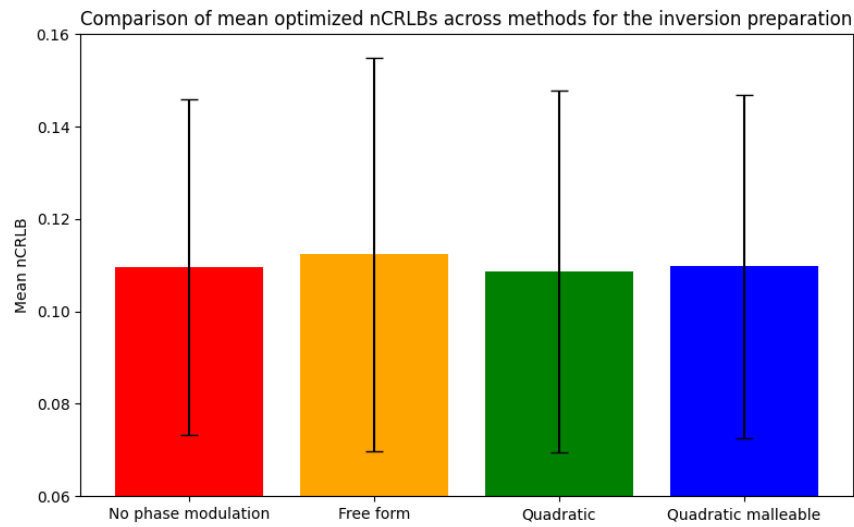


Figure 5.6: Results of the optimization process of the flip and phase trains of the optional inversion pulse preparation.

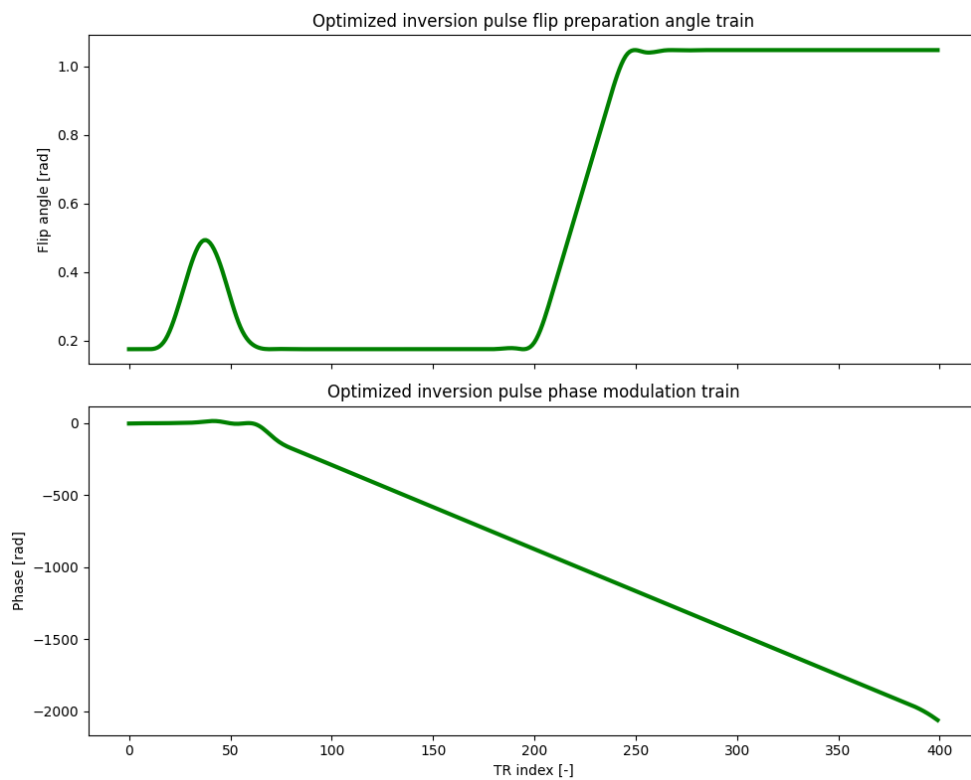
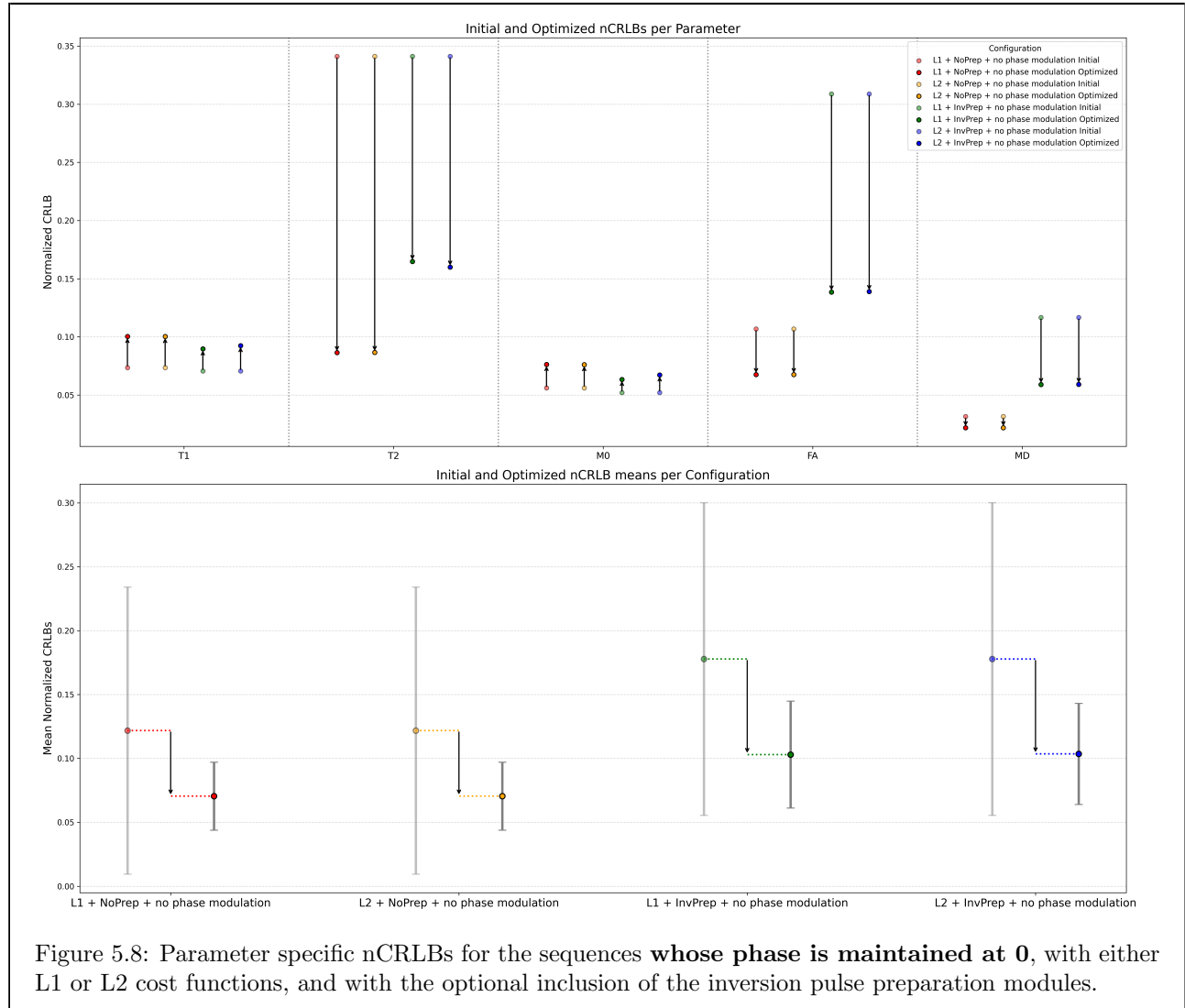


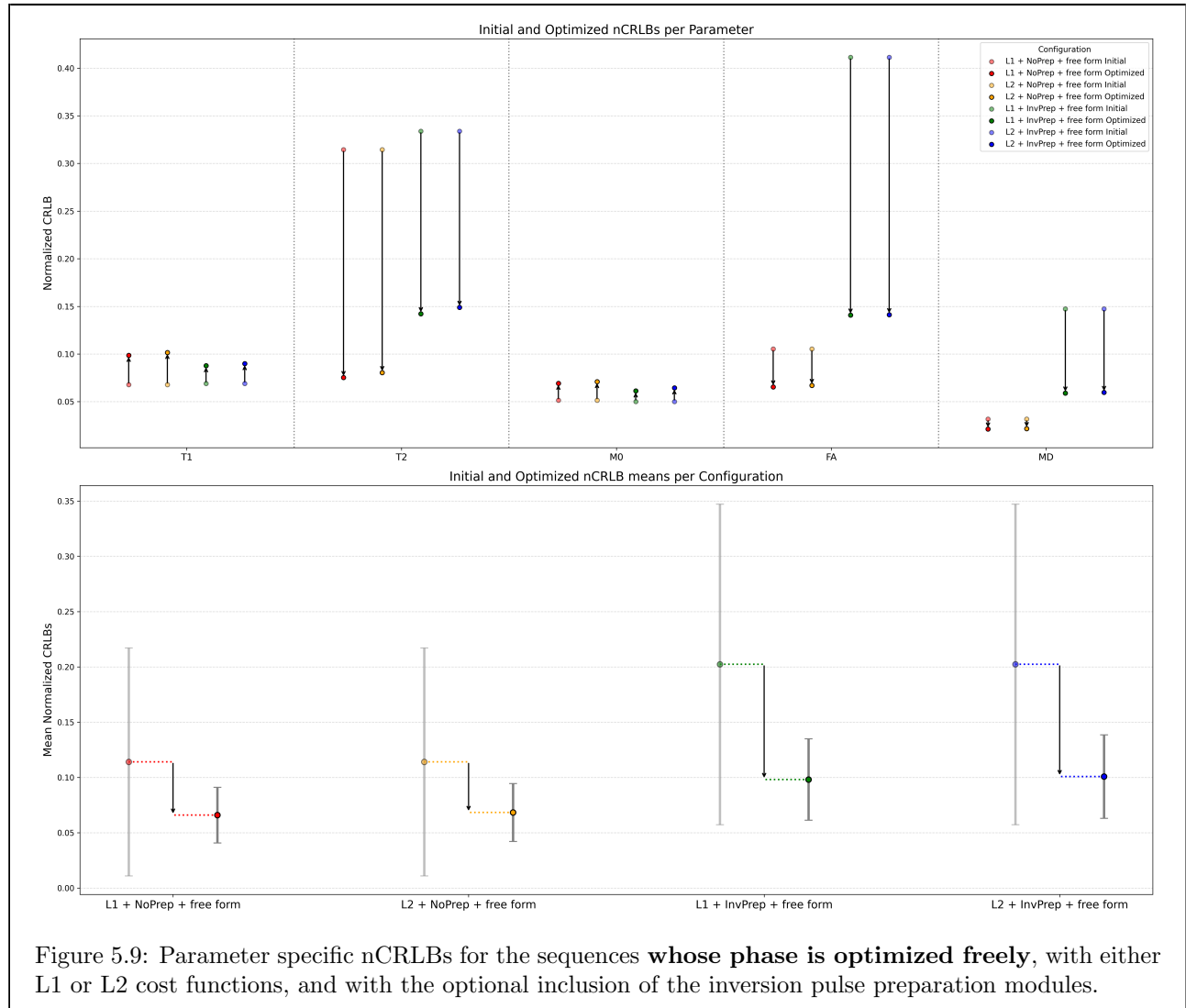
Figure 5.7: Optimized flip and phase trains for the 400 TR-long optional inversion pulse preparation module.

the standard deviation associated to it. The latter is especially important as it testifies about the homogenisation of the nCRLBs by the optimization process.

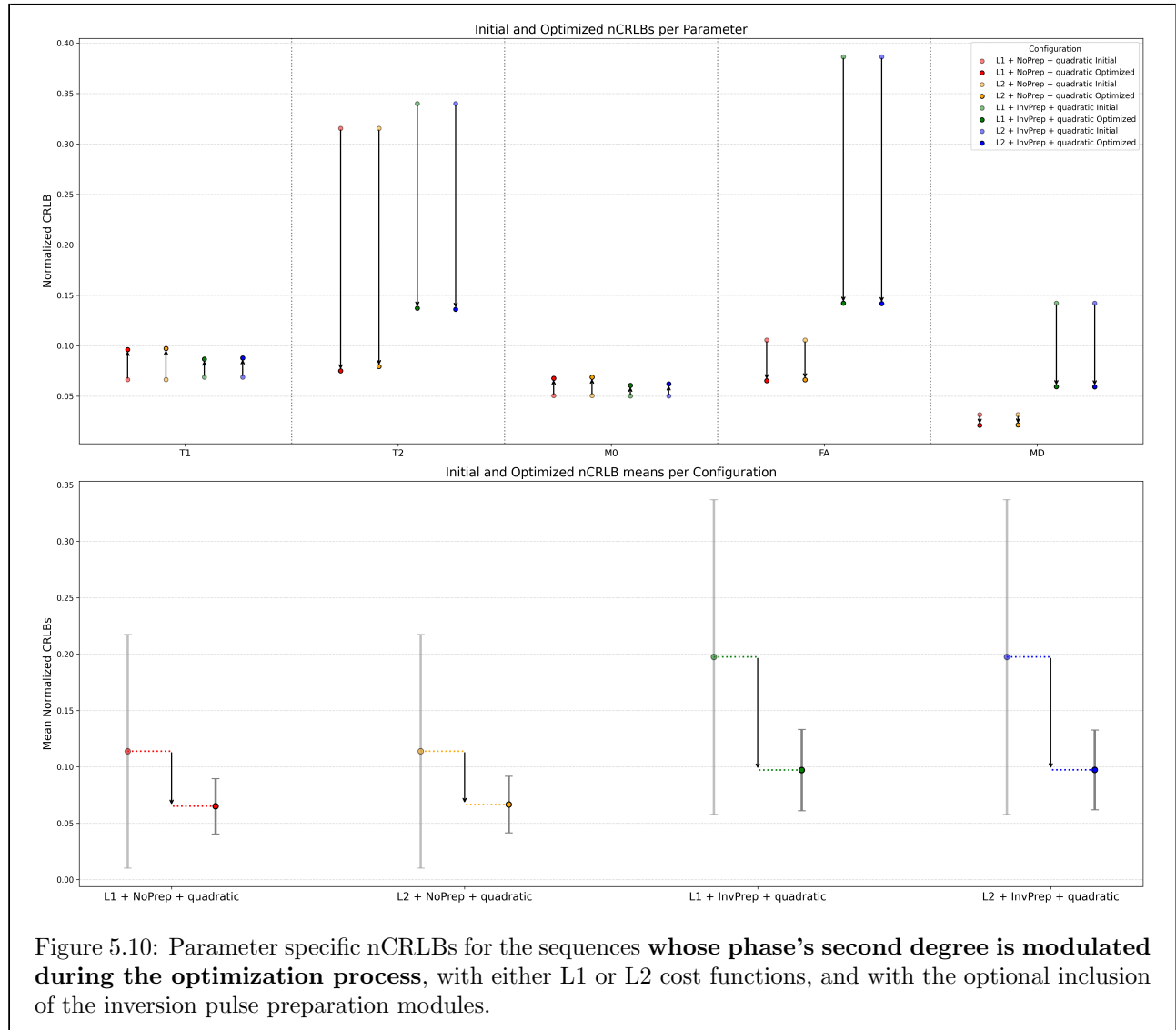
### Method 1: Constant Phase



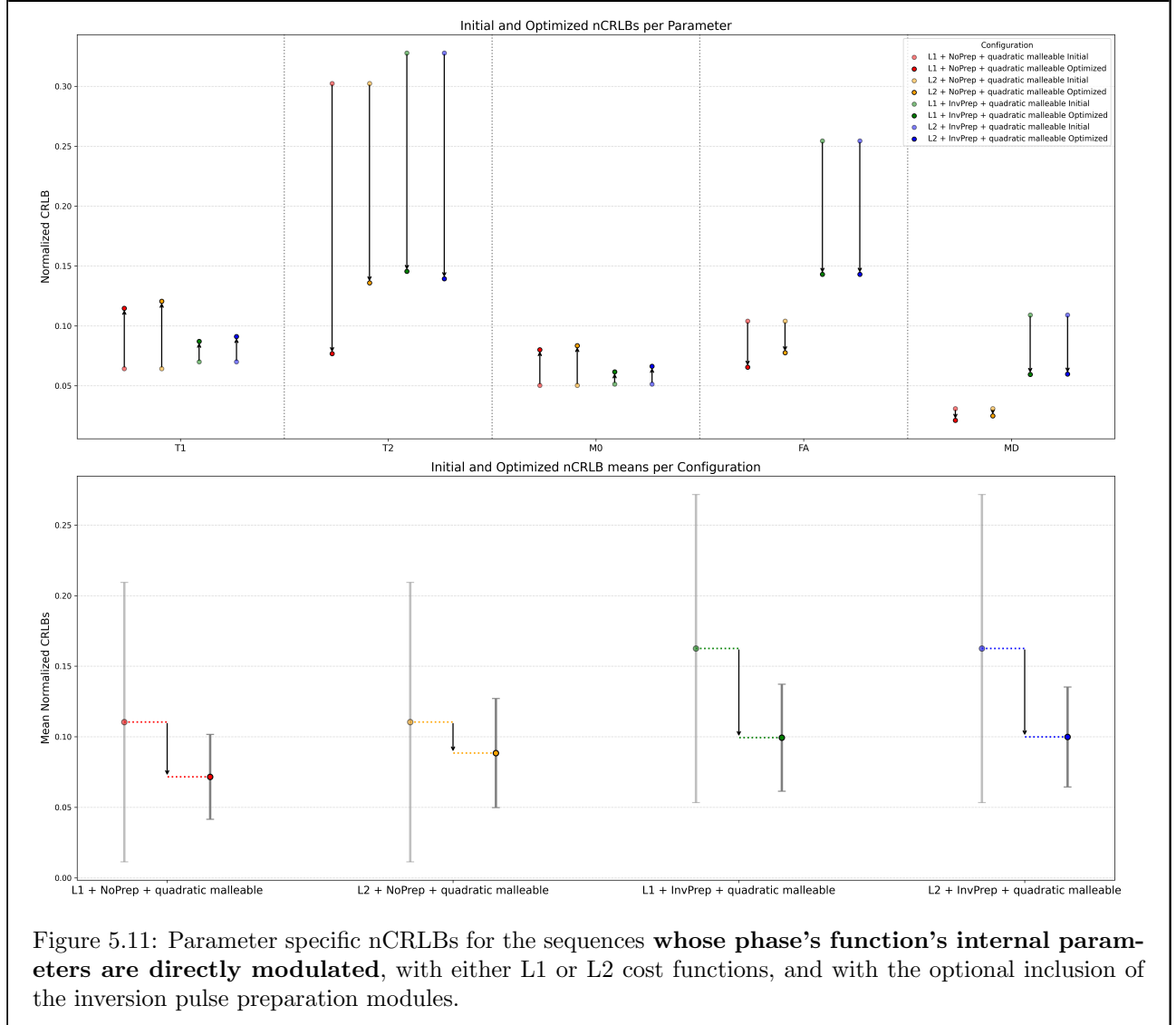
## Method 2: free form phase modulation optimization



## Method 3: Second order modulation



## Method 4: Modulation parameter dependent function



## 5.5 Identification of the optimal sequence type

**5.5.1 Optimal flip angle and phase trains** As shown in subsection 5.4.2 and further illustrated in Figure 5.12, the seemingly best performing sequence was the one obtained with an **L1**-type cost function, **without** the addition of the optional inversion pulse preparation module in the start of the sequence and with the **second-order** phase modulation method (in the figures, labelled as *Quadratic*). The sequence outperforms the other sequences both when it comes to the mean normalized Cramér-Rao lower bound and the standard-deviation in the latter. The numerical results for the nCRLBs of the best sequence are displayed in Table 5.1.

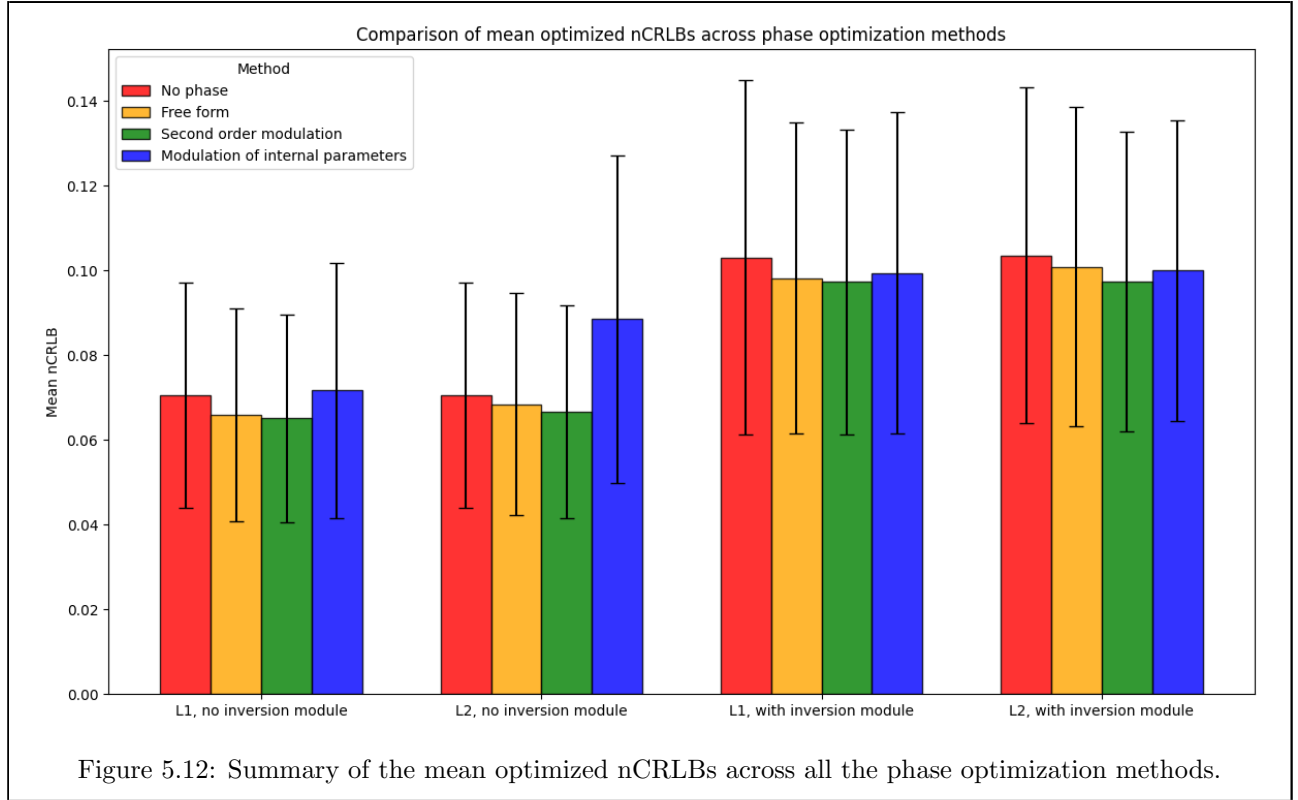


Table 5.1: Normalized Cramér-Rao lower bounds for the optimal sequence using the best phase modulation method, sequence type and cost function order.

| Parameter          | nCRLB  |
|--------------------|--------|
| $T_1$              | 0.1147 |
| $T_2$              | 0.0768 |
| $M_0$              | 0.0802 |
| $FA$               | 0.0655 |
| $MD$               | 0.0212 |
| Mean               | 0.0651 |
| Standard deviation | 0.0245 |

The flip angle and phase trains are displayed in Figure 5.13. In order to maximize tissue contrast (i.e. lower the nCRLBs of the different parameters in those tissues), the optimizer seems to have "chosen" a flip angle configuration maximizing the information content of the signal resulting from the diffusion encoding. Due to the limited lifetime of the latter, the flip angles seem to sharply decrease after about 70 TRs, and bottom out at the minimum value of  $\frac{\pi}{180}$ . At TR index 150, the flip angles seem to increase again, most probably to reintroduce some signal variation. The phase train does not resemble its initial configuration at all, with the optimization "erasing" the discontinuity that was typical to the starting phase train. The phase seems to evolve at a close to linear rate through the train, with the exception that at the end of the train, the phase increase seems to be tapered.

**5.5.2 Magnetization signal evolution obtained using optimal sequences** Using the optimized flip angle and phase modulation trains, the magnetization signal evolution over the entire sequence can be computed. An example of such a signal trajectory is shown in Figure 5.14. It can be observed that the diffusion weighting is lost rapidly, but that diffusion-related contrast is gradually re-established towards the end of the sequence.

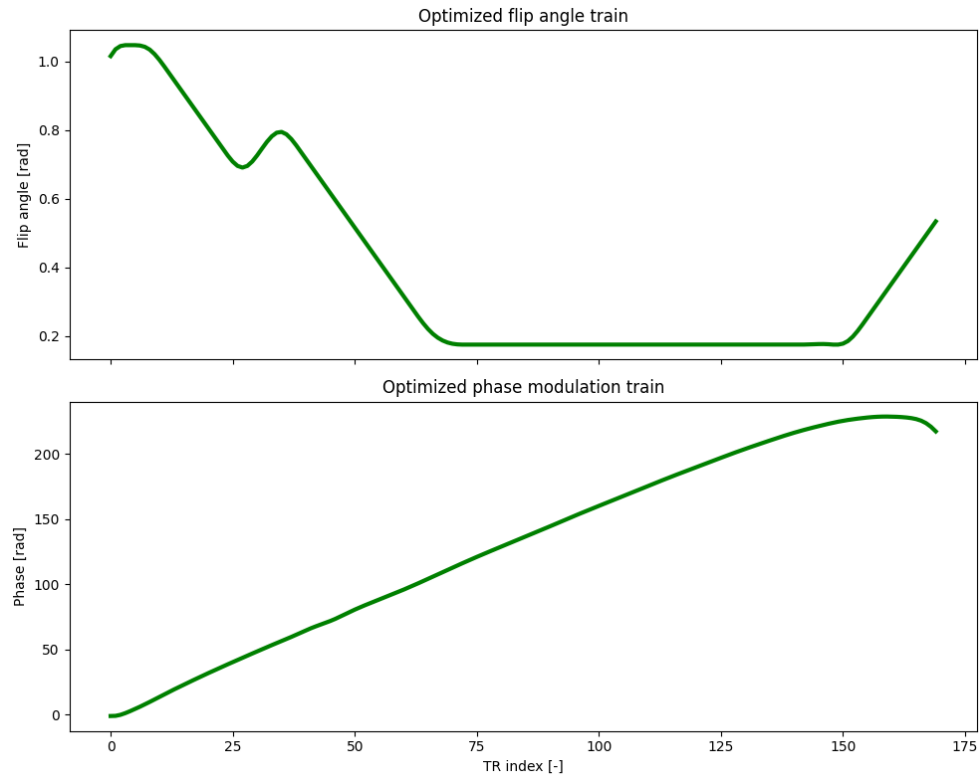
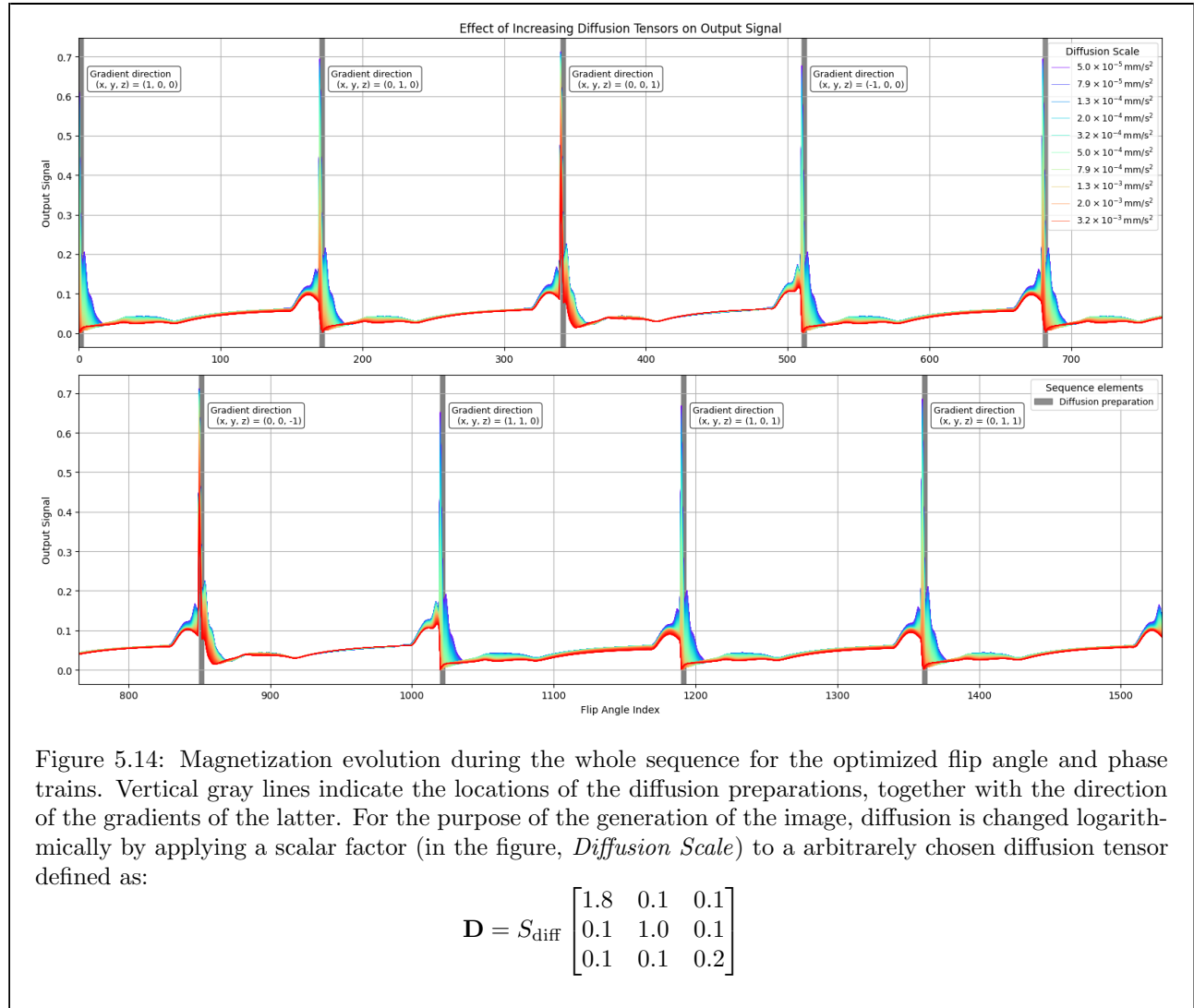


Figure 5.13: Optimized flip and phase trains for the 170 TR-long modules subsequent to diffusion preparation modules.



## 5.6 Results summary

| Component                                          | Details                                                           | Options / Notes                                                                             |
|----------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>1. EPG States</b>                               |                                                                   |                                                                                             |
| Number of States                                   | 5 states chosen                                                   | Stabilization of nCRLB beyond 5; balances accuracy and speed                                |
| <b>2. Initial Trains</b>                           |                                                                   |                                                                                             |
| Phase Train                                        | Optimized internal parameter for initial phase                    | B-spline parametrized with 30 knots; discontinuity causes largest error                     |
| Flip Angle Train                                   | Sliced MRF flip angle train used as initialization                | Smooth B-spline fit with minimal error                                                      |
| <b>3. B-spline Knots</b>                           |                                                                   |                                                                                             |
| Number of Knots                                    | 30 knots sufficient                                               | Parameter-specific nCRLBs stabilize; reduces dimensionality                                 |
| <b>4. Inversion Pulse Preparation Optimization</b> |                                                                   |                                                                                             |
| Optimization Result                                | Slight advantage for second-order (quadratic) phase modulation    | Flip angle train shows two-stage behavior: transient low-flip followed by high-flip plateau |
| <b>5. Phase Modulation Methods Evaluation</b>      |                                                                   |                                                                                             |
| Methods Compared                                   | Constant phase, free form, 2nd order, parametric function         | 16 combinations tested (phase method $\times$ inversion prep $\times$ cost order)           |
| Best Method                                        | 2nd order phase modulation + L1 cost + no inversion prep          | Lowest mean nCRLB and best homogenization                                                   |
| <b>6. Optimal Sequence Characteristics</b>         |                                                                   |                                                                                             |
| Flip Angle Train                                   | Sharp decrease after 70 TR, minimum plateau, increase near TR 150 | Reflects balance between diffusion signal lifetime and information content                  |
| Phase Train                                        | Nearly linear increase, smooth, no discontinuities                | Phase evolution tapered towards the end                                                     |
| nCRLBs (mean/std)                                  | 0.0651 / 0.0245                                                   | Parameters: $T_1$ , $T_2$ , $M_0$ , $FA$ , $MD$                                             |

## 6 Discussion

This study focused on the design and optimization of a magnetic resonance fingerprinting (MRF) acquisition scheme through simultaneous modulation of the flip angle and phase trains. The primary objective was to assess how such modulation strategies influence the encoding efficiency for both relaxation and diffusion parameters, with an emphasis on improving estimation precision.

The results clearly demonstrate that incorporating phase modulation, particularly second-order phase variation, leads to a systematic reduction in the mean normalized Cramér-Rao lower bound (nCRLB) across all examined parameters. In addition, this approach yields a more uniform nCRLB distribution, as evidenced by a decrease in its standard deviation. Together, these findings point to a promising avenue for enhancing the performance of diffusion-weighted MRF protocols through targeted phase and flip angle design.

In the following paragraphs, these outcomes are examined in more detail, considering their implications in the context of previous work, the methodological trade-offs involved, and the limitations encountered during the study.

The study of the effect of the number of used EPG states corroborates the claim that employing reduced EPG state matrices yields outcomes comparable to those using larger matrices. This reduction was also employed by Lee et al. [46], where a slightly larger number of 10 states was used. Many simulations [54] in the field of MRF employ matrix definitions of large dimensions (in the range of  $3 \times 150$ ), however even when simulating simple, non-diffusion-weighted sequences. This study demonstrates that the simulation and the optimization of even more complicated and irregular sequences is plausible while making use of smaller EPG matrices ( $3 \times 150$ ).

An expected limitation of this study is in the choice of the functions used for the determination of the initial phase modulation. Instead of optimizing every function with regards to all the parameters that define it (for example, for a sine function, its phase, frequency and amplitude), a single parameter was chosen for every function. While this does indeed restrict the solution space and the "potential" of some functions, it stabilizes the optimization and prevents the search from becoming intractable.

The use of the B-spline proves itself to be useful to reduce the dimensionality of the optimization problem. The B-spline has inherent continuous properties, making the optimization problem easier. However, this continuity also imposes structure on the solution space. Specifically, the B-spline representation excludes highly irregular or non-smooth solutions, such as those with abrupt ridges or sharp transitions. As a result, the solution space becomes constrained, potentially excluding certain classes of non-smooth yet locally optimal configurations. In summary, the parametrization is a trade-off between computational ease and solution space size.

The normalized Cramér-Rao lower bounds values must be interpreted with care. The weighing vector  $w$  in Equation 4.1 was assigned a constant values 1 to each of its components. This is not the case in other studies, where  $w$  occasionally serves as a compression factor [63] for the normalized Cramér-Rao lower bounds, resulting in seemingly smaller, hence better, values. Furthermore, the choice of the specified noise  $\sigma^2$  level may vary over different MRF implementations. As can be observed in Equation 4.1, the value of a certain nCRLB is linearly dependent on  $\sigma^2$ , highlighting its direct impact on the order of the obtained results. The value employed in this study was the same one as the one used by Zhao et al [63] in a similar experiment (but without the inclusion of diffusion and phase modulation in the optimization process). Observe that the results are comparable across all the optimization schemes.

Though not impacting the order of the numerical value directly, there is a number of choices that directly strongly impact the optimization results. The study employs equal weights (values all set to 1 in  $w^{(2)}$  present in Equation 4.2) for every considered tissue. The choice of the properties of the latter were made such as to simulate an equal presence of white matter, grey matter, and cerebrospinal fluid.

One would might have expected that the replacement of 2 diffusion modules with an inversion preparation module would have resulted in better sequences. As demonstrated in subsection 5.4.2, this is not actually the case. Figure 5.8, Figure 5.9, Figure 5.10 and Figure 5.11 yield the exact opposite conclusion. A decrease in the diffusion Cramér-Rao lower bounds is expected, as the inversion pulse preparation module does not feature

diffusion encoding. The increase that is actually observed might be cost to pay for the decrease of the  $T_1$  and, most importantly,  $T_2$  nCRLBs. A possible explanation may be that the increase in the diffusion parameters also directly propagates to the relaxation parameters.

## 7 Conclusion

This thesis serves as a presentation of a comprehensive framework for the robust optimization of diffusion-enhanced MRF sequences, with particular emphasis on the joint optimization of flip angle and phase modulation trains. Building on the extended phase graph formalism, B-spline parameterization, and information-theoretic optimization criteria based on the Cramér–Rao lower bound, the study addresses the challenge of improving parameter estimation precision for both relaxation and diffusion properties in a simultaneous manner.

The results demonstrate several key findings. First, the number of extended phase graph states required for accurate simulation and optimization are shown to exhibit a rapid convergence, enabling computationally efficient modeling without the compromise of accuracy. Secondly, the B-spline parametrization of flip angle and phase trains effectively reduced dimensionality while ensuring smoothness and stability of the optimization process, albeit at the cost of excluding highly irregular solutions. Third, systematic evaluation of different phase modulation strategies revealed that second-order (quadratic) phase modulation, in combination with an L1-type cost function and without the addition of an inversion preparation module, yielded the most favorable trade-off across all parameters. This optimal configuration achieved the lowest mean normalized CRLBs (6.51%) and the best homogenization across  $T_1$ ,  $T_2$ ,  $M_0$ ,  $FA$ , and  $MD$  (2.45%), thereby improving the robustness of parameter encoding.

Importantly, the study also revealed that the anticipated benefits of introducing an inversion preparation block did not materialize. Instead, its inclusion reduced signal diversity and weakened encoding efficiency, particularly for relaxation parameters. This finding underscores the sensitivity of MRF optimization to sequence design choices and highlights the importance of systematic, data-driven evaluation over intuitive assumptions.

Taken together, these contributions provide evidence that principled, information-driven optimization of both flip angle and phase modulation substantially enhances the encoding power of diffusion-MRF sequences. Beyond methodological value, the presented framework lays the groundwork for clinically relevant protocols capable of simultaneously mapping relaxation and diffusion parameters within a single acquisition. Such protocols would not only reduce scan time but also improve diagnostic value by consolidating multi-contrast imaging into a unified workflow.

Future work should extend the current framework to in vivo validation as this study lacks experiments conducted in real-life situations, explore adaptive weighting schemes reflecting clinical priorities, and investigate robustness against confounding effects such as  $B_0/B_1$  inhomogeneities and physiological noise. By bridging advanced sequence design with quantitative imaging needs, this work contributes a step toward the broader goal of making quantitative MRI, and specifically MRF, a reliable and practical tool in clinical practice.

# A Appendix

## A.1 Validation via Numerical Phantom Reconstruction

The following methods were originally presented by Heesterbeek et al.[15] and were extended in this work to accommodate diffusion-weighted MRF sequences. However, due to time constraints, no conclusive results were obtained from this extension during the course of the study. To validate the effectiveness of optimized sequence designs, a comprehensive simulation study using synthetic phantoms can be performed. These phantoms allow controlled testing of the quantitative mapping pipeline, including signal simulation, image reconstruction, and parameter estimation via dictionary matching. The validation aims to assess whether optimized sequences enable accurate recovery of tissue parameters, particularly longitudinal relaxation time ( $T_1$ ), transverse relaxation time ( $T_2$ ), mean diffusivity (MD), and fractional anisotropy (FA).

**A.1.1 Dictionary Construction** A four-dimensional dictionary  $\mathcal{D}$  is generated by simulating the signal evolution for a wide range of tissue parameter combinations. Specifically, the following ranges are used:

- $T_1 \in [150, 3000]$  ms, logarithmically spaced (50 values)
- $T_2 \in [30, 1000]$  ms, logarithmically spaced (50 values)
- MD scaling factor  $\in [0, 1]$  (50 values)
- FA scaling factor  $\in [0, 1]$  (50 values)

For each parameter combination ( $T_1, T_2, \text{MD}, \text{FA}$ ), the signal evolution  $\mathbf{m}(t)$  is simulated using the Extended Phase Graph (EPG) formalism, incorporating diffusion via a symmetric diffusion tensor  $\mathbf{D} \in \mathbb{R}^{3 \times 3}$ . The base diffusion tensor was defined as:

$$\mathbf{D}_0 = 10^{-3} \begin{bmatrix} 1 & 0.1 & 0.1 \\ 0.1 & 1 & 0.1 \\ 0.1 & 0.1 & 1 \end{bmatrix} [\text{mm}^2/\text{s}]$$

This tensor is systematically modulated to vary MD and FA. Given a mean diffusivity scaling factor  $s_{\text{MD}}$  and a fractional anisotropy scaling factor  $s_{\text{FA}}$ , the modified tensor  $\mathbf{D}$  is constructed by adjusting the diagonal entries along the  $yy$  and  $zz$  axes to create anisotropic variants:

$$\begin{aligned} \mathbf{D}_{yy} &\leftarrow \mathbf{D}_{yy} \cdot (0.1 + 0.9 \cdot s_{\text{FA}}), & \mathbf{D}_{zz} &\leftarrow \mathbf{D}_{zz} \cdot (0.1 + 0.9 \cdot s_{\text{FA}}) \\ \mathbf{D} &\leftarrow s_{\text{MD}} \cdot \mathbf{D} \end{aligned}$$

The dictionary signals are simulated using the optimized flip angle train  $\{\alpha_n\}_{n=1}^N$  and the corresponding quadratic phase modulation  $\{\phi_n\}_{n=1}^N$ . Each signal vector  $\mathbf{m}_\theta \in \mathbb{C}^N$  in the dictionary corresponds to a specific parameter tuple  $\theta = (T_1, T_2, \text{MD}, \text{FA})$ , where  $N$  is the total number of timepoints.

**A.1.2 Phantom Generation** To test the robustness of the sequence across a range of parameter combinations, we construct a two-dimensional digital phantom of size  $121 \times 121$  voxels with a checkerboard pattern. The phantom alternates between two tissue types:

- White squares:  $T_1 = 750$  ms,  $T_2 = 70$  ms
- Gray squares:  $T_1 = 1250$  ms,  $T_2 = 90$  ms

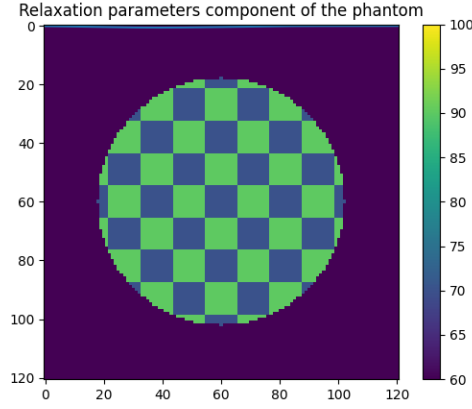


Figure A.1: The **relaxation parameters component** of the phantom. The two different  $T_1$  &  $T_2$  combinations are alternated following a chessboard-like pattern.

A circular mask with a relative radius of  $r = 0.35 \times 121 = 42$  voxels is applied to restrict the region of interest and simulate anatomical boundaries. The relaxation parameters component of the phantom is displayed in Figure A.1.

For the diffusion component, a second phantom is constructed using a tile-based approach with varying combinations of MD and FA, ranging from low anisotropy/high diffusivity to high anisotropy/low diffusivity. This generates a spatially structured map of tensors  $\{\mathbf{D}_{i,j}\}_{i,j}$  from which ground-truth maps of MD and FA can be computed:

$$\text{MD}(\mathbf{D}) = \frac{1}{3} \text{Tr}(\mathbf{D}), \quad \text{FA}(\mathbf{D}) = \sqrt{\frac{1}{2} \left( 3 - \frac{1}{\text{Tr}(\mathbf{R}^2)} \right)}, \quad \text{where } \mathbf{R} = \frac{\mathbf{D}}{\text{Tr}(\mathbf{D})}$$

The diffusion components of the phantom are displayed in Figure A.2.

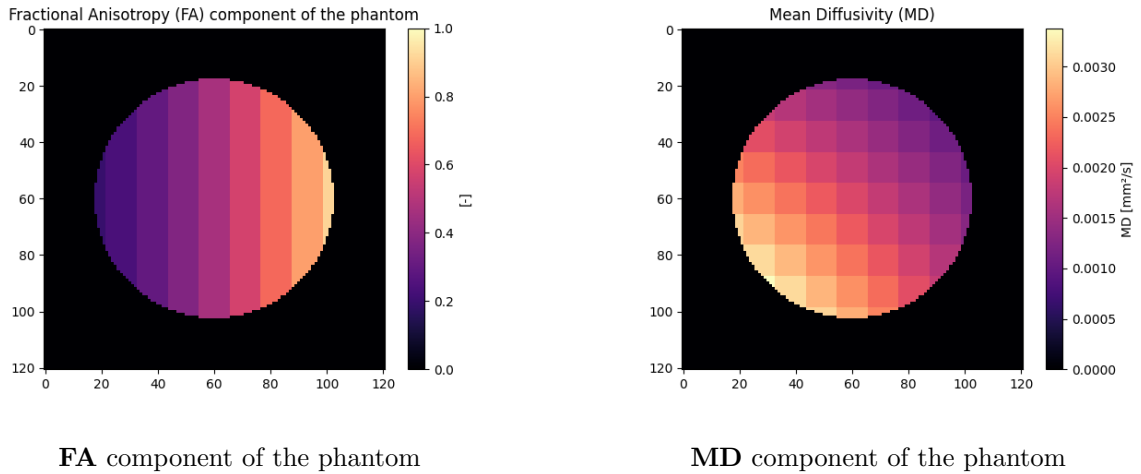


Figure A.2: Diffusion components of the phantom used to study the quality of the reconstruction process using the optimized sequence.

**A.1.3 Signal Forward Simulation and Image Reconstruction** For each voxel  $(i, j)$ , the magnetization time course  $\mathbf{m}_{i,j} \in \mathbb{C}^N$  is synthesized as a linear combination of precomputed white and gray

matter signals, weighted according to the phantom structure. A complex-valued quadratic phase field  $\Phi(x, y)$  is applied to simulate realistic spatial phase variation:

$$\mathbf{m}_{i,j}^{(\text{final})} = \rho_{i,j} \cdot \mathbf{m}_{i,j} \cdot e^{i\Phi(i,j)}$$

Here,  $\rho_{i,j}$  is the proton density (binary mask from the circular support). Optional additive complex Gaussian noise is added to simulate stochastic SNR levels.

The forward-simulated time series images are then undersampled using realistic spiral k-space trajectories. The spiral sampling pattern is either golden-angle based or derived from vendor-specific spiral interleaves. For each timepoint  $t_n$ , non-uniform Fourier encoding is applied via a NUFFT operator  $\mathcal{F}_n^{\text{spiral}}$ , and the corresponding k-space signal  $s_n$  is computed:

$$s_n = \mathcal{F}_n^{\text{spiral}} [\mathbf{m}_n^{(\text{final})}]$$

The images are then reconstructed by applying the adjoint NUFFT operator, including density compensation to correct for non-uniform sampling:

$$\hat{\mathbf{m}}_n = (\mathcal{F}_n^{\text{spiral}})^H [s_n \cdot \text{DCF}_n]$$

**A.1.4 Parameter Estimation via Dictionary Matching** At each voxel  $(i, j)$ , the measured signal vector  $\hat{\mathbf{m}}_{i,j} \in \mathbb{C}^N$  is normalized and matched to the dictionary via inner product maximization:

$$\theta_{i,j}^* = \arg \max_{\theta \in \mathcal{D}} |\langle \hat{\mathbf{m}}_{i,j}, \mathbf{m}_\theta \rangle|$$

The estimated amplitude  $M_0$  is computed from the signal norm:

$$M_0^{(i,j)} = \frac{\|\hat{\mathbf{m}}_{i,j}\|_2}{\|\mathbf{m}_{\theta_{i,j}^*}\|_2}$$

The phase of the complex inner product provides the estimated phase offset at that voxel.

Finally, the estimated MD and FA values are post-processed by finding the closest diffusion tensor in the precomputed grid of tensors  $\{\mathbf{D}_{k,\ell}\}$  using Euclidean distance in the (MD, FA) parameter space.

**A.1.5 Validation Outputs** The reconstructed parameter maps,  $T_1(x, y)$ ,  $T_2(x, y)$ , MD( $x, y$ ), and FA( $x, y$ ), are then visualized and compared to their known ground-truth counterparts derived from the phantom definitions. This validation demonstrates the ability of the optimized sequences and reconstruction pipeline to recover spatially structured relaxation and diffusion properties under realistic encoding conditions.

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